PART: Baking Attributes into High-Dimensional Vectors for Filtered Approximate Nearest Neighbor Search

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ABSTRACT

Filtered Approximate Nearest Neighbor Search (FANNS), which retrieves similar high-dimensional vectors subject to attribute constraints, is critical for numerous applications. However, existing FANNS methods suffer substantial search performance degradation under low pass rate conditions, meaning only a tiny fraction of the dataset satisfies the constraints. This is mainly because prior methods cannot unify attributes and vectors in a shared space: prioritizing attributes induces disconnected sub-indexes, hindering efficient search traversal, whereas prioritizing vectors prevents valid candidates from clustering, forcing the algorithm to exhaust its search budget on invalid candidates. Motivated by this, we propose PART, which natively bakes attributes into the vector space via an attribute-aware rotation over raw vectors. This rotation preserves proximity among vectors with identical attributes while pushing vectors of different attributes toward near-orthogonality. PART provides theoretical error bounds to guarantee accurate distance computation on these attribute-baked vectors. Furthermore, we extend PART to support complex predicates, including multi-attribute AND, numeric range filters, and hybrid workloads. Finally, PART is integrated into mainstream ANN indexes, including RaBitQ quantization, IVF, and graph indexes. Comprehensive experiments validate that PART achieves a favorable Pareto frontier between search efficiency and accuracy across all datasets. Under an extreme low pass rate of 0.1% and a target recall of 0.95, PART achieves up to a 5.5× speedup over the state-of-the-art baseline.

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The source code, data, and/or other artifacts have been made available at https://github.com/Lsyhprum/PART.

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1 INTRODUCTION

Unstructured data objects such as images, text, and videos are typically mapped to *high-dimensional vectors* encoding their semantic content. These vectors are then queried using Approximate Nearest Neighbor Search (ANNS) to return the top-k objects similar to a given query [23, 36, 45, 55]. While ANNS has been widely applied in content retrieval [16, 18], recommendation systems [6, 58], and Retrieval-Augmented Generation (RAG) [26, 39], modern applications increasingly rely on structured *attribute* constraints to enhance the quality of semantic query results. For example, an enterprise RAG system may retrieve relevant design images belonging to a specific department, or an AI agent might search for similar financial reports published within a specific time window. Such queries require retrieving the top-k objects that are most semantically similar (e.g., in visual or textual proximity) among those satisfying attribute constraints (e.g., departmental categories or temporal bounds). To meet this practical demand, Filtered Approximate Nearest Neighbor Search (FANNS) has emerged as an important problem and attracted growing attention [1, 4, 30, 39, 56].

Limitations of Existing Methods. Many FANNS methods have been proposed with the aim of pushing the Pareto frontier of search efficiency and accuracy [28, 49]. A key obstacle lies in handling variable *pass rate*, the fraction of vectors satisfying filter conditions [22]. Early paradigms including Pre-filtering and Post-filtering rely on attribute-vector decoupled execution strategies [47]: Pre-filtering resolves the attribute constraints before vector search, but often degenerates into an expensive brute-force scan under high pass rates [41]. Conversely, Post-filtering retrieves candidates first and discards invalid matches later, wasting massive distance computations when valid candidates are sparse under low pass rates [28]. These paradigms fail to evaluate attributes and vectors jointly, rendering them highly fragile across a wide spectrum of pass rates.

Recently, Joint-filtering has established itself as the mainstream FANNS paradigm. This paradigm is typically built upon graph-based indexes [36, 50], achieving FANNS either by restructuring *attribute-dedicated* indexes during the construction phase (e.g., FDiskANN [13], iRangeGraph [54]), or by fusing attribute matching into the routing logic of an *attribute-agnostic* index during the search phase (e.g., ACORN [41], NaviX [42]). However, Joint-filtering struggles to maintain the robust connectivity crucial for efficient graph traversal under low pass rates. This is fundamentally

Table 1: Comparison of PART against existing Joint-filtering methods. More ★'s indicate better.

	Ours (PART)	Attribute-dedicated (e.g., FDiskANN)	Attribute-agnostic (e.g., ACORN)
Attribute Format	A sequence of random phase angles.	Implicit in graph topology.	External structured metadata.
Filtering Strategy	Transforms vectors via phase-based rotation.	Restructures index topology.	Attribute-aware routing strategy.
Predicate Support	Common predicates (`EQ`, `AND`, range, hybrid).	Single predicate.	Arbitrary predicates.
Index Compatibility	Highly compatible with VQ, IVF, and graph.	Requires designing custom structures.	Requires modifying search algorithms.
Pass Rate Adaptability	Preserves nearest-neighbors proximity and attribute distinctiveness. ★★★	Fragmented topology lacking cross-attribute proximity edges. ★	Topologically disconnected candidates waste distance computations. ★

because these methods still treat the attribute space and vector space as distinct entities, failing to natively unify them. Consequently, for attribute-dedicated methods, the partitioned sub-indexes struggle to reliably establish proximity edges across different attribute values. Conversely, for attribute-agnostic methods, the attribute matched candidates remain topologically disconnected, forcing the search to waste massive distance computations on invalid neighbors.

Our Solution PART. To overcome the pass rate limitations of existing methods by breaking this fundamental structural tradeoff, we propose PART, a Phase-Based Attribute Rotation Transform method that natively bakes structured attributes into a continuous vector space while simultaneously maintaining precise evaluations of vector similarity and attribute matching with theoretical guarantees. Its key insight is encoding attribute values into distinct random *phase angles*, which are subsequently utilized to drive an *attribute-aware rotation* on the vector space. Vectors sharing identical attribute values undergo the same angular transformation to preserve their original spatial proximity, thereby ensuring the accuracy of vector similarity results. Conversely, vectors with differing attribute values are separated by a sufficiently large angular margin, guaranteeing the soundness of attribute matching. By unifying attributes and vectors at the representation level, PART allows mainstream ANN indexes to inherently support FANNS without requiring ad-hoc modifications to their core graph topology or routing logic.

PART achieves this transformation through a two-step pipeline: **(1)** PART constructs a high-dimensional composite phase angle vector for each attribute value, where each element in the angle vector represents a 2D coordinate plane. This is achieved by repeatedly evaluating a dedicated-seed random hash function across the $B = d/2$ disjoint dimension pairs. **(2)** PART translates each angle from the angle vector into a 2×2 standard rotation block for its respective dimension pair, assembling them into a d-dimensional block-diagonal orthogonal matrix. By applying this matrix, the attribute value is baked directly into the data vector. The orthogonal nature of this transformation preserves the similarity between vectors sharing identical attribute values. Furthermore, by leveraging extensive theoretical tools related to the Johnson-Lindenstrauss Lemma [9, 40], we prove that the similarity between vectors with mismatched attribute values tightly concentrates around zero.

We further extend the basic PART to support complex predicates, including multi-attribute `AND`, numeric ranges, and hybrid workloads that combine discrete attributes with numeric ranges. Specifically, for *multi-attribute* `AND`, it superimposes the phase rotations of all associated attributes. For *numeric ranges*, it anchors the baseline rotation at the range center, translating numerical deviations into proportional angular shifts to guarantee distance preservation for valid targets and attenuation for boundary violations. Ultimately, for *hybrid workloads*, PART seamlessly unifies discrete and numeric filters within the same space, leveraging its highly compositional phase rotation framework.

Finally, because this transformation operates entirely at the data representation level, PART serves as a highly compatible module. We integrate it with mainstream ANNS methods, including RaBitQ [12], IVF, and graph-based indexes, by operating directly on the transformed vectors without altering their core search logic or topological structures. Furthermore, by deterministically parameterizing the transformation of each attribute value with its own dedicated pseudo-random seed, PART incurs *negligible storage overhead* and rapidly reconstructs the rotation matrices on-the-fly.

We evaluate PART extensively on standard and real-world datasets, empirically validating three core advantages. **(1)** Compared with state-of-the-art baselines, PART achieves up to 5.5× higher QPS at a pass rate of 0.1%, and maintains a 1.5× to 3× advantage at a high pass rate of 50%. Furthermore, it sustains up to a 2.5× speedup on ArXiv dataset, which features a highly *skewed* real-world attribute distribution. **(2)** PART robustly maintains a strict 90% Recall@100 target even under highly constrained filtering spaces, such as an extreme 0.01% pass rate or ultra-large attribute cardinalities of up to 10,000. **(3)** Our empirical results on real-world datasets align tightly with our established theoretical bounds, rigorously confirming the reliability and completeness of our theoretical analysis.

Contributions. We summarize our contributions as follows:

- We identify that mainstream Joint-filtering FANNS methods fail to natively unify structured attribute and continuous vector spaces, inherently causing connectivity bottlenecks that severely degrade search performance under low pass rates.
- We propose PART, a novel method that bakes attributes into a continuous vector space, enabling accurate evaluation of attribute matching and vector similarity through a unified computation.
- We extend PART to support complex predicates, including multi-attribute `AND`, numeric ranges, and hybrid workloads.
- We provide theoretical guarantees establishing the attribute matching soundness and vector similarity completeness for both single-attribute filtering and multi-attribute `AND`. Additionally, we derive error bounds for numeric ranges and hybrid filtering.

2 PRELIMINARIES

We formalize Approximate Nearest Neighbor Search (ANNS) and its filtered variant FANNS, which restricts search to vectors satisfying attribute constraints. Table 2 summarizes frequently used notation.

2.1 Approximate Nearest Neighbor Search

Given a dataset $\mathcal{D} \subset \mathbb{R}^d$ of N data vectors in a d-dimensional space and a query vector $\mathbf{q}_r \in \mathbb{R}^d$, Nearest Neighbor Search (NNS) returns the ground-truth set $\mathcal{G}(\mathbf{q}_r) \subset \mathcal{D}$ of the k data vectors

Table 2: Frequently used notation.

Notation	Definition
$\mathcal{D}$	the vector dataset
$\mathbf{o}_r$, $\mathbf{q}_r$	the raw data and query vector
$\mathbf{o}$, $\mathbf{q}$	the normalized data and query vector
$\mathcal{A}$	the attribute schema
$\mathbf{a}$, $\mathbf{t}$	the attribute set of data vector and query target
a, t	the attribute value and query target value
$\mathbf{p}_a$	the phase angle vector of attribute value a
$\hat{\mathbf{o}}$, $\hat{\mathbf{q}}$	the attribute-baked data vector and query vector

most similar to $\mathbf{q}_r$ under a given distance metric. Since exact search incurs prohibitive computational cost at scale, Approximate Nearest Neighbor Search (ANNS) is widely used in practice. ANNS retrieves a candidate set $\mathcal{R}(\mathbf{q}_r)$ of k approximate nearest neighbors, trading a minor drop in search quality for substantially higher efficiency. ANNS quality is measured by Recall, which is formally defined as:

$$\text{Recall@}k(\mathbf{q}_r) = \frac{|\mathcal{G}(\mathbf{q}_r) \cap \mathcal{R}(\mathbf{q}_r)|}{k} \quad , \tag{1}$$

while the search efficiency of ANNS is typically measured by query throughput, i.e., Queries Per Second (QPS).

ANNS methods are commonly classified by their underlying structure into four families: tree- [7, 34, 38], hash- [14, 19, 29], IVF- [3, 5, 24], and graph-based methods [10, 11, 36]. The latter two dominate industrial deployments [2, 39, 48]. Both approaches evaluate distances on full-precision vectors, which constitutes the main computational cost at large scale. To mitigate this cost, Vector Quantization (VQ) is widely coupled with IVF and graphs for faster search and smaller memory footprint. For example, FAISS [23] uses coarse clusters to prune the search space, relying on PQ [24] to accelerate distance computations within the target clusters. DiskANN [45] compresses vectors into PQ codes to fit in memory, fetching raw vectors from an SSD only for reranking. SymphonyQG [15] embeds low-bit quantizers [12] into the graph traversal process, thereby bypassing the overhead of full-precision distance evaluations.

2.2 Filtered ANNS

In modern real-world applications, vectors are widely accompanied by structured metadata. Suppose that each raw data vector $\mathbf{o}_r \in \mathcal{D}$ is associated with attributes $\mathbf{a}$ (e.g., $\mathbf{a} = \{\text{color} : \text{“blue”},\ \text{price} : 85\}$) drawn from a schema $\mathcal{A}$. Filtered Approximate Nearest Neighbor Search (FANNS) addresses retrieval over such data by formulating a query as a pair $(\mathbf{q}_r, \phi)$, coupling the raw query vector $\mathbf{q}_r$ with a boolean predicate ϕ over $\mathcal{A}$ (e.g., color = “blue” `AND` price < 100). This predicate defines the target subset of eligible candidates $\mathcal{D}_\phi = \{\mathbf{o}_r \in \mathcal{D} \mid \phi(\mathbf{a}) = \text{true}\}$. The fraction of these eligible candidates is defined as the pass rate of predicate ϕ, denoted as:

$$\text{pass rate} = \frac{|\mathcal{D}_\phi|}{|\mathcal{D}|} \quad . \tag{2}$$

When $|\mathcal{D}_\phi| \approx k$ or smaller, the search must recover nearly all eligible candidates, making the problem hardest. Since pass rates drastically impact query performance [28], numerous FANNS methods have been proposed, which can be broadly categorized into three paradigms based on the execution order of vector search and attribute filtering: Pre-filtering [52], Post-filtering, and Joint-filtering. Pre-filtering first materializes the matched attributes as a bitmap or ID-list, followed by a brute-force vector search over that subset. While straightforward, it *suffers severely under high pass rates* [28, 33], where the subset of eligible candidates $\mathcal{D}_\phi$ approaches the full dataset $\mathcal{D}$. Post-filtering retrieves approximate nearest neighbors first and subsequently discards those failing the predicate. However, this strategy becomes *prohibitively expensive under low pass rates*, as the search must retrieve a massive volume of candidates to yield enough valid matches [28].

Joint-filtering has emerged as the mainstream paradigm, interleaving attribute evaluation directly with ANNS index traversal. It splits into two families, attribute-dedicated and attribute-agnostic, depending on when the attribute is reconciled with the vector space:

- Attribute-dedicated methods [49] bake the predicate into index topology at construction time, restructuring the graph into attribute-scoped sub-indexes. For example, FDiskANN [13] builds a subgraph for every label and stitches them into a unified index, iRangeGraph [54] targets numeric ranges by partitioning the attribute domain into a binary search tree of segments, building an independent subgraph per segment. However, this partitioning fails to establish reliable proximity edges across subgraphs, triggering routing failures under low pass rates.
- Attribute-agnostic methods [42, 43] instead maintain a predicate-blind index and inject the filter dynamically during search, enforcing explicit per-candidate checks over an otherwise ordinary graph. For example, ACORN [41] builds a standard HNSW graph and traverses it at query time. To maintain navigability across sparse valid candidates, it performs run-time multi-hop expansion, actively utilizing non-matching nodes as routing bridges to reach the next valid region. However, relying on these invalid bridge nodes inevitably wastes massive distance computations, severely degrading query efficiency under low pass rates.

In summary, the fundamental mismatch between continuous vectors and discrete attributes inherently disrupts the graph connectivity of existing FANNS methods under low pass rates. Lacking theoretical bounds that natively unify attribute filtering and vector distance estimation, these heuristic-driven methods suffer unpredictable performance degradation across varying pass rates. *This motivates us to design a novel FANNS method that adapts to diverse pass rates, underpinned by theoretical guarantees for exact attribute matching and accurate distance estimation.*

3 THE BASIC PART METHOD

In this section, we introduce the Phase-based Attribute Rotation Transform (PART), a novel FANNS method providing theoretical guarantees to overcome the unpredictable performance of existing approaches across varying pass rates. To clearly articulate the core mechanism, we first formalize the basic PART method assuming a *single* categorical attribute with an *exact-match* predicate. The extension of this foundational method to support multi-attribute filtering and complex predicates is detailed in §4.

3.1 Baking Attributes into Vectors

As summarized in Algorithm 1, consider a vector database $\mathcal{D} \subset \mathbb{R}^d$ where each data vector $\mathbf{o}$ is associated with an attribute value a from an attribute schema $\mathcal{A}$. We assume both data and query vectors

are *normalized* (i.e., $\|\mathbf{o}\|_2 = \|\mathbf{q}\|_2 = 1$). PART generates attribute-baked vectors that jointly capture vector similarity and attribute matching through a two-step pipeline: **(1)** Phase-Based Attribute Encoding: Translates the attribute value into a sequence of pseudo-random, unique phase angles. **(2)** Attribute-Baking Vector Rotation: Assembles these phase angles into an orthogonal matrix and applies it to rotate the vector $\mathbf{o}$, ultimately embedding the attribute value directly into the continuous vector space. We detail two steps below.

Phase-Based Attribute Encoding. Unlike existing FANNS methods that rely on decoupled data structures such as bitmaps or ID-lists to encode attributes, PART translates discrete attribute values into continuous phase angles. First, PART partitions the d vector dimensions into $B = d/2$ disjoint dimension pairs[1] (line 1 in Algorithm 1). Second, PART assigns each pair $j \in \{1, \ldots, B\}$ a salt s_j drawn independently from a common uniform distribution (line 2). The attribute value a is then mapped to a specific phase angle for the j-th pair using a pseudo-random hash function:

$$\psi_j(a) = 2\pi \cdot \mathrm{rand}(a, s_j) \in [0, 2\pi) \quad , \tag{3}$$

where $\mathrm{rand}(\cdot,\cdot)$ is a deterministic hash function that, over the random choice of salt s_j, outputs a value uniformly distributed in $[0, 1)$ (lines 6-8). Finally, by evaluating this hash function across all B pairs, PART concatenates the results to form a high-dimensional composite phase angle vector $\mathbf{p}$:

$$\mathbf{p}_a = \big(\psi_1(a), \ldots, \psi_B(a)\big)^\top \in [0, 2\pi)^B \quad . \tag{4}$$

This phase angle vector $\mathbf{p}_a$ serves as the unique representation for the attribute value. The set of random salts $\{s_j\}_{j=1}^B$ is sampled once offline during system initialization and shared globally across all data and query encodings. This mapping guarantees that identical attribute values evaluated under the same global salts deterministically map to the exact same high-dimensional angle vector, paving the way for the subsequent vector rotation.

Attribute-Baking Vector Rotation. PART translates the phase angle vector $\mathbf{p}$ into an orthogonal transformation matrix. Each phase angle $\psi_j(a)$ parameterizes a 2×2 standard rotation matrix for its respective j-th dimension pair. Collectively, these blocks are concatenated via the matrix direct sum ($\bigoplus$) to form a d-dimensional, block-diagonal phase rotation matrix $\Phi(a) \in \mathbb{R}^{d \times d}$:

$$\Phi(a) = \bigoplus_{j=1}^{B} \mathbf{R}(\psi_j(a)), \quad \text{where} \quad \mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad . \tag{5}$$

Because every block is a standard two-dimensional rotation, the assembled matrix $\Phi(a)$ is strictly orthogonal. To bake the attribute value a into the data vector $\mathbf{o}$, PART simply applies this orthogonal rotation matrix to the vector: $\hat{\mathbf{o}} = \Phi(a)\mathbf{o}$ (line 12).

During the search phase, a normalized query vector $\mathbf{q}$ targeting attribute value t undergoes the exact same transformation: $\hat{\mathbf{q}} = \Phi(t)\mathbf{q}$. By applying these rotations, vectors equipped with identical attribute values ($a = t$) maintain their original spatial alignment, whereas vectors with differing attribute values ($a \neq t$) are forced into nearly orthogonal orientations. Since the relative rotation $\Phi(t)^\top\Phi(a)$ is governed by the per-pair angle differences $\psi_j(a) - \psi_j(t)$, which are random when $a \neq t$, the baked inner product (IP) concentrates near zero.

[1]Modern embedding models mostly produce even-dimensional vectors (e.g., Gemini [25] and Qwen3 [59]). PART thus assumes an even d. An odd d can be trivially padded with a zero without affecting the distance semantics.

Algorithm 1 Basic PART Method

Require: normalized vector database $\mathcal{D} \subset \mathbb{R}^d$, attribute schema $\mathcal{A}$
Ensure: attribute-baked vector database $\hat{\mathcal{D}}$
1: $B \leftarrow d/2$; ▷ Number of disjoint dimension pairs
2: Initialize global random salts $\{s_1, \ldots, s_B\}$;
3: $\hat{\mathcal{D}} \leftarrow \emptyset$;
4: **for each** vector $\mathbf{o} \in \mathcal{D}$, associated attribute value $a \in \mathcal{A}$ **do**
5: // Step 1: Phase-Based Encoding
6: **for** $j = 1$ **to** B **do**
7: $\psi_j(a) \leftarrow 2\pi \cdot \mathrm{rand}(a, s_j)$;
8: **end for**
9: $\mathbf{p} \leftarrow \big(\psi_1(a), \ldots, \psi_B(a)\big)^\top$; ▷ Composite phase angle vector
10: // Step 2: Attribute-Baking Rotation
11: $\Phi(a) \leftarrow \bigoplus_{j=1}^{B} \mathbf{R}(\psi_j(a))$; ▷ Assemble orthogonal matrix
12: $\hat{\mathbf{o}} \leftarrow \Phi(a)\mathbf{o}$; $\hat{\mathcal{D}} \leftarrow \hat{\mathcal{D}} \cup \hat{\mathbf{o}}$;
13: **end for**
14: **return** $\hat{\mathcal{D}}$

Example 1. *Consider a database with dimension $d = 4$. Let $\mathbf{o}_1 = \mathbf{o}_2 = (0.5, 0.5, 0.5, 0.5)^\top$ be two identical normalized vectors, where $\mathbf{o}_1$ is associated with attribute value $a_1 = 42$ and $\mathbf{o}_2$ with $a_2 = 99$. First, via Phase-Based Attribute Encoding, evaluating $a_1 = 42$ yields phase angles $\psi_1(42) = \pi/2$ and $\psi_2(42) = \pi$ for the two dimension pairs. These natively form the composite phase angle vector $\mathbf{p}_{a_1} = (\pi/2, \pi)^\top$. Through Attribute-Baking Vector Rotation, $\mathbf{p}_{a_1}$ parameterizes the orthogonal matrix $\Phi(a_1)$ as a direct sum of two 2×2 blocks:*

$$\Phi(a_1) = \mathbf{R}(\pi/2) \oplus \mathbf{R}(\pi) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad .$$

Conversely, the deterministic hash for $a_2 = 99$ yields a distinct angle vector $\mathbf{p}_{a_2} = (\pi, \pi/2)^\top$, assembling an independent rotation matrix:

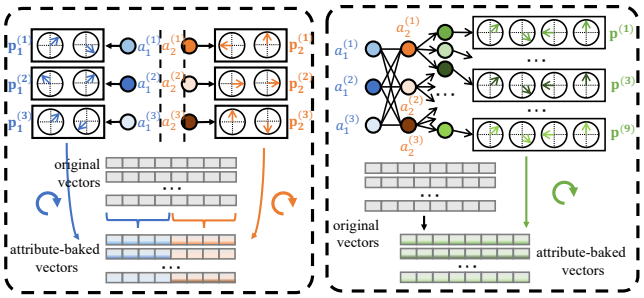
$$\Phi(a_2) = \mathbf{R}(\pi) \oplus \mathbf{R}(\pi/2) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad .$$

The attribute values are then baked into data vectors: $\hat{\mathbf{o}}_1 = \Phi(a_1)\mathbf{o}_1 = (-0.5, 0.5, -0.5, -0.5)^\top$, and $\hat{\mathbf{o}}_2 = \Phi(a_2)\mathbf{o}_2 = (-0.5, -0.5, -0.5, 0.5)^\top$.

Given a normalized query $\mathbf{q} = (0.5, 0.5, 0.5, 0.5)^\top$ targeting attribute value $t = a_1 = 42$, it undergoes the exact same transformation $\hat{\mathbf{q}} = \Phi(a_1)\mathbf{q} = (-0.5, 0.5, -0.5, -0.5)^\top$. (1) For the matched vector $\mathbf{o}_1$ ($a_1 = t$), the relative rotation is $\Phi(a_1)^\top\Phi(a_1) = \mathbf{I}$. Their original IP is strictly preserved: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_1 \rangle = 1$. (2) For the mismatched vector $\mathbf{o}_2$ ($a_2 \neq t$), the relative rotation $\Phi(a_1)^\top\Phi(a_2)$ alters the spatial alignment. $\hat{\mathbf{q}}$ and $\hat{\mathbf{o}}_2$ are forced into orthogonality, penalizing their IP: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_2 \rangle = 0$, effectively pruning $\mathbf{o}_2$.

3.2 Effectiveness Theoretical Analysis

This section establishes the theoretical foundation explaining why PART is effective for FANNS. Specifically, we formalize its filtering mechanism through two guarantees: *attribute matching soundness* and *vector similarity completeness*. Respectively, they ensure the structural pruning of attribute-mismatched vectors, and the authentic distance evaluation for identically attributed vectors.



(a) *Algorithm I: CPR* (b) *Algorithm II: APR*

Figure 1: Illustration of CPR & APR for multi-attribute AND.

Theorem 1 (Attribute Matching Soundness). *Given a data vector* $\mathbf{o}$ *associated with a scalar attribute* a*, and a query vector* $\mathbf{q}$ *targeting attribute* t*, assume data vectors and queries are normalized. If* $a \neq t$*, the baked IP has zero mean,* $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$*, with a variance bounded by* $\mathrm{Var}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 1/d$*. The tail probability that the mismatch IP exceeds a similarity threshold* $\epsilon > 0$ *decays exponentially with dimensionality* d*:* $\mathbb{P}\Big(\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle \geq \epsilon\Big) \leq \exp\left(-\frac{d \cdot \epsilon^2}{4}\right)$.

Proof. Let $B = d/2$ and $\Delta_j = \psi_j(a) - \psi_j(t)$. The baked IP decomposes as $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \sum_{j=1}^{B} X_j$ with $X_j = u_j \cos\Delta_j + v_j \sin\Delta_j$, where $u_j = \langle \mathbf{q}^{(j)}, \mathbf{o}^{(j)} \rangle$ and $v_j = \langle \mathbf{q}^{(j)}, J\mathbf{o}^{(j)} \rangle$ (J the 90° rotation). Since the hash is pairwise-independent, Δ_j is uniform on $[0, 2\pi)$, so $\mathbb{E}[\cos\Delta_j] = \mathbb{E}[\sin\Delta_j] = 0$, $\mathrm{Var}(\cos\Delta_j) = \mathrm{Var}(\sin\Delta_j) = 1/2$, and cross-terms vanish; hence $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$. By the 2D identity $\langle \mathbf{x}, \mathbf{y} \rangle^2 + \langle \mathbf{x}, J\mathbf{y} \rangle^2 = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$, we have $u_j^2 + v_j^2 = \|\mathbf{q}^{(j)}\|^2 \|\mathbf{o}^{(j)}\|^2 \triangleq w_j$, so $\mathrm{Var} = \frac{1}{2} \sum_{j=1}^{B} w_j$. Under balanced per-subspace variance, $w_j = 1/B^2$, giving $\mathrm{Var} = \frac{1}{2}(B \cdot 1/B^2) = 1/d$. Finally, $X_j \in [-\sqrt{w_j}, \sqrt{w_j}]$ since $w_j = u_j^2 + v_j^2$. Hoeffding's inequality then yields $\mathbb{P}(\sum_j X_j \geq \epsilon) \leq \exp\big(-2\epsilon^2 / \sum_j (2\sqrt{w_j})^2\big) = \exp\big(-2\epsilon^2/(4\sum_j w_j)\big)$. With $\sum_j w_j = B \cdot 1/B^2 = 1/B$ and $B = d/2$, this simplifies to $\exp(-d\epsilon^2/4)$. □

Consequently, the mismatch IP is strictly bounded by $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = O(1/\sqrt{d})$ with high probability, ensuring that false positives are structurally suppressed to a negligible residual error.

Theorem 2 (Vector Similarity Completeness). *If a normalized data vector* $\mathbf{o}$ *matches the query attribute* t *(i.e.,* $a = t$*), then their baked IP strictly equals the original IP:* $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$.

Proof. $a = t \Rightarrow \Phi(a) = \Phi(t)$. By definition, the baked IP is $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = (\Phi(t)\mathbf{q})^\top (\Phi(a)\mathbf{o}) = \mathbf{q}^\top \Phi(t)^\top \Phi(a)\mathbf{o}$. Since $\Phi(t)$ is an orthogonal rotation matrix, $\Phi(t)^\top \Phi(t) = \mathbf{I}$. Thus, $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \mathbf{q}^\top \mathbf{I}\mathbf{o} = \langle \mathbf{q}, \mathbf{o} \rangle$. For any two matched vectors, both $\hat{\mathbf{o}}_1$ and $\hat{\mathbf{o}}_2$ are obtained by applying the same orthogonal transform $\Phi(t)$. Since orthogonal transforms preserve IPs and distances, the order of $\langle \hat{\mathbf{q}}, \cdot \rangle$ over the matched set exactly equals the order of $\langle \mathbf{q}, \cdot \rangle$. □

4 THE EXTENSION OF PART

In this section, we extend PART beyond single-attribute equality filtering to support complex predicates, including multi-attribute AND, continuous numeric ranges, and arbitrary mixtures of both.

4.1 Multi-Attribute AND Extension

We design two baking algorithms to support AND predicates within PART. We first describe an intuitive block-partitioning solution (*Algorithm I*), and then present a global-superposition mechanism (*Algorithm II*) that provides rigorous theoretical guarantees.

Algorithm I: Composite Phase Rotation (CPR). CPR partitions the vector space into multiple sub-blocks, applying single-attribute rotations to each sub-block. This approach decomposes a multi-attribute AND into a series of localized, single-attribute filters. As shown in Figure 1(a), for a data vector tagged with an M-attribute tuple $\mathbf{a} = (a_1, \ldots, a_M)$, CPR evenly partitions the $d/2$ dimension-pairs into M disjoint blocks $\{\mathcal{B}_1, \ldots, \mathcal{B}_M\}$. CPR independently maps each attribute value a_m to its designated block $\mathcal{B}_m$ to derive a localized angle vector $\mathbf{p}_{a_m}$, where $\psi_j(a_m)$ extends Eq. 3 to the m-th attribute value. These sub-rotations are then assembled into a unified block-diagonal matrix: $\Phi(\mathbf{a}) = \bigoplus_{m=1}^{M} \Big(\bigoplus_{j \in \mathcal{B}_m} \mathbf{R}(\psi_j(a_m)) \Big)$. By applying the exact same partitioning to the query vector targeting $\mathbf{t} = (t_1, \ldots, t_M)$, a data vector fully aligns with the query if and only if all attributes match ($a_m = t_m$ for all m), organically embedding AND into the vector space.

Example 2. *Consider a* $d = 8$ *space where we execute a multi-attribute* AND *query targeting* $M = 2$ *attributes. CPR partitions the 4 dimension-pairs into* $\mathcal{B}_1$ *(pairs 1-2) for the first attribute and* $\mathcal{B}_2$ *(pairs 3-4) for the second. For a data vector* $\mathbf{o}$ *tagged with attribute tuple* $\mathbf{a} = (42, 99)$ *(i.e.,* $a_1 = 42, a_2 = 99$*), its localized phase signatures (e.g.,* $\mathbf{p}_{a_1} = (\pi/2, \pi)$ *and* $\mathbf{p}_{a_2} = (\pi, \pi/2)$*) directly assemble the unified rotation matrix:* $\Phi(\mathbf{a}) = \mathbf{R}(\pi/2) \oplus \mathbf{R}(\pi) \oplus \mathbf{R}(\pi) \oplus \mathbf{R}(\pi/2)$*. The vector is then baked:* $\hat{\mathbf{o}} = \Phi(\mathbf{a})\mathbf{o}$*. Given a query vector* $\mathbf{q}$ *where* $\langle \mathbf{q}, \mathbf{o} \rangle = 1$*, when* AND *is satisfied, querying* $\mathbf{t} = (42, 99)$ *applies identical rotations across all blocks, fully preserving the score* $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$*. Conversely, when* AND *fails, querying* $\mathbf{t}' = (42, 77)$ *aligns only on* $\mathcal{B}_1$.

However, CPR compromises soundness (Theorem 1) due to dimensionality dilution. While Theorem 1 suppresses false positives via the exponential bound $\exp(-d \cdot \epsilon^2/4)$ over the full dimensionality d, CPR partitions the space and reduces the effective dimensionality of each block to d/M. This structural division creates a vulnerability for near-miss vectors (e.g., matching $M - 1$ attributes but failing on exactly one). In such cases, orthogonal penalization occurs *only* within the single mismatched block, degrading the mismatch error bound from $O(1/\sqrt{d})$ to $O(\sqrt{M/d})$. Meanwhile, the mismatched block, suffering from reduced dimensionality, fails to generate sufficient orthogonal penalty to push the overall score below the similarity threshold. This leads to an increase in false positives for AND queries with a large number of attributes.

Algorithm II: Additive Phase Rotation (APR). APR superimposes the phase angles of all M attributes over *every* 2D subspace. As shown in Figure 1(b), for a data vector labeled with an M-attribute tuple $\mathbf{a} = (a_1, \ldots, a_M)$, APR aggregates the per-attribute phases, each modulated by an independent salt $s_m(j)$, to compute a phase angle for each dimension pair j. Here $\psi_{m,j}(a_m) = 2\pi \cdot \mathrm{rand}(a_m, s_m(j))$ extends Equation 3 to the m-th attribute, and the salts $\{s_m(j)\}$ are drawn i.i.d. across both m and j, independent of the base salts $\{s_j\}$. The rotation matrix is then constructed:

$$\Phi(\mathbf{a}) = \bigoplus_{j=1}^{d/2} \mathbf{R}\Big(\sum_{m=1}^{M} s_m(j) \psi_{m,j}(a_m) \Big) \quad . \tag{6}$$

By applying the exact same superposition to the query targeting $\mathbf{t} = (t_1, \ldots, t_M)$, let $\delta_{m,j} = \psi_{m,j}(a_m) - \psi_{m,j}(t_m)$ denote the phase

difference for the m-th attribute on pair j. The total relative phase shift between the baked data vector and the query strictly evaluates to $\Delta_j = \sum_{m=1}^{M} s_m(j)\delta_{m,j}$.

Example 3. *Revisit the $d = 8$ space with an $M = 2$ attribute query. APR superimposes both attributes across* all *4 pairs. For a data vector* $\mathbf{o}$ *tagged with attribute tuple* $\mathbf{a} = (42, 99)$ *(i.e., $a_1 = 42, a_2 = 99$), the angle on any pair $j \in \{1, 2, 3, 4\}$ is a salted sum of both phases: $s_1(j)\psi_{1,j}(a_1) + s_2(j)\psi_{2,j}(a_2)$. Given a query* $\mathbf{q}$ *where $\langle \mathbf{q}, \mathbf{o} \rangle = 1$, when* AND *is satisfied, querying* $\mathbf{t} = (42, 99)$ *yields zero phase differences ($\delta_{1,j} = \delta_{2,j} = 0$) globally, fully preserving the score $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$. Conversely, when* AND *fails, querying $\mathbf{t}' = (42, 77)$ matches the first attribute but generates a random non-zero difference $\delta_{2,j}$ for the second. Because APR superimposes phases, the total shift $\Delta_j = s_2(j)\delta_{2,j} \neq 0$ affects* every *pair. Moreover, since $\delta_{2,j}$ varies randomly across j, these shifts are independent across pairs, thereby scrambling the entire vector space simultaneously upon any single mismatch.*

By superimposing the rotations of all M attributes across *every* dimension-pair j, APR eliminates the dimensionality dilution vulnerability inherent in CPR. Concretely, when $\mathbf{a} = \mathbf{t}$ all per-attribute differences $\delta_{m,j}$ vanish, so $\Delta_j = 0$ for every pair and the IP is preserved exactly; when some attribute mismatches, the resulting Δ_j remains a random phase over all $B = d/2$ pairs, so the concentration argument of Theorem 1 applies over the full dimensionality d rather than d/M. We therefore adopt APR as the foundational mechanism for subsequent extensions.

4.2 Continuous Numeric Range Extension

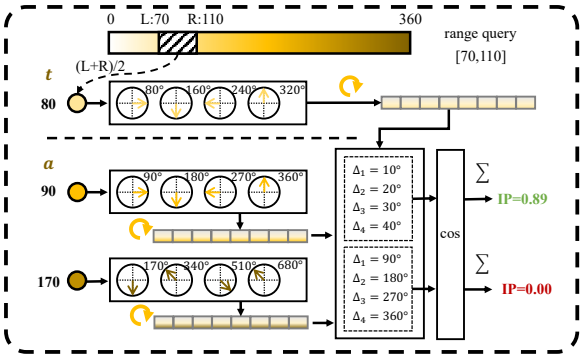
Unlike exact-match predicates that rely on random hashing to penalize arbitrary mismatches, range filtering needs to preserve numerical locality to construct a valid bounding window. To this end, we next introduce Scaled Phase Rotation (SPR).

Algorithm III: Scaled Phase Rotation (SPR). Unlike exact-match predicates, which rely on the random phase $\psi_j(a)$ to penalize arbitrary mismatches, range filtering must preserve numerical locality so that nearby values remain similar. SPR anchors the baseline rotation at the range center $t = (l + u)/2$ and translates deviations from t into proportional angular shifts: values *inside* $[l, u]$ undergo bounded rotations that preserve similarity, whereas values *outside* accumulate destructive phase offsets. Specifically, for a numeric attribute $a \in [0, L]$ (L its value range), the range phase on the j-th dimension-pair ($j \in \{1, \ldots, B\}$, $B = d/2$) is

$$\phi_j(a) = \pi \frac{j}{B} \frac{a}{L} \quad . \tag{7}$$

The relative shift against t is $\Delta_j = \phi_j(a) - \phi_j(t) = \pi \frac{j}{B} \frac{a-t}{L}$. For $a \in [l, u]$, $|\Delta_j| \leq \pi \frac{u-l}{2L}$, which is small when $u - l \ll L$, so $\cos \Delta_j \approx 1$ and the inner product is largely preserved; as a moves away from t, Δ_j grows linearly with $|a - t|$ and with j/B, so higher-index pairs yield larger shifts and out-of-window vectors are progressively down-ranked. When $u - l$ is not small relative to L, the query partitions $[l, u]$ into $m = \lceil (u - l)/w \rceil$ sub-queries of width $w \ll L$ and merges their candidate sets.

Example 4. *Consider a $d = 8$ space ($B = 4$ pairs) with value range $[0, 100]$, a range query $[40, 60]$ centered at $t = 50$, and three data vectors $\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3$ with $\langle \mathbf{q}, \mathbf{o}_i \rangle = 1$. Assuming even variance across the 4 pairs, the baked score is $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle = \frac{1}{4} \sum_{j=1}^{4} \cos \Delta_j$. For the center $\mathbf{o}_1$ ($a_1 = 50$), $\Delta_j = 0$ for all j, so the score stays 1. For an in-window boundary vector $\mathbf{o}_2$ ($a_2 = 60$), the offset 10 gives $\Delta_j = \pi \frac{j}{40}$ and a score ≈ 0.98, slightly attenuated but safely retrieved. For an out-of-window vector $\mathbf{o}_3$ ($a_3 = 90$), the offset 40 gives $\Delta_j = \pi \frac{j}{10}$ and $\Delta_4 = 0.4\pi$, lowering the score to ≈ 0.66 — clearly below the in-window case, though not collapsed to zero.*



Algorithm III: SPR

Figure 2: Illustration of SPR for range attribute.

SPR's primary theoretical advantage is *bounded IP preservation*, providing provable in-window fidelity over unpredictable heuristic degradation. We formalize this via two bounds. **(1)** For a single query (m=1), the maximum phase shift is bounded by $\pi w/(2L)$, so the in-window IP deviation is $O((w/L)^2)$. For narrow ranges (e.g., $w/L \leq 3\%$), the resulting attenuation is below 0.2%, generating high-fidelity candidates for a lightweight rerank. **(2)** For wider ranges, *sub-query partitioning* tightens the bound: executing m equal-width sub-queries compresses the maximum phase shift to $\pi w/(2mL)$, so the in-window IP converges to the exact semantic IP as m grows. By scaling m with the window width to bound the residual error, SPR preserves the scores of in-window members, enabling the reranker to recover the exact top-k given a high-recall ANN index.

4.3 Hybrid Predicates Extension

Real-world applications frequently demand the simultaneous evaluation of discrete categories and continuous numerical ranges. To fuse these heterogeneous filtering within a single space, we introduce Hybrid Additive Phase Rotation (HAPR).

Algorithm IV: Hybrid Additive Phase Rotation (HAPR). Built upon the additive phase superposition paradigm, HAPR integrates APR and SPR into every 2D subspace, reusing the discrete salts $s_m(j)$ of Equation 6 and the range phase of Equation 7 without introducing new parameters. Specifically, consider a hybrid query comprising M discrete attributes $\mathbf{t}^{\text{disc}} = (t_1^{\text{disc}}, \ldots, t_M^{\text{disc}})$ and C range attributes $\{[l_1, u_1], \ldots, [l_C, u_C]\}$, where L_k is the value range of the k-th numeric attribute and $t_k^{\text{num}} = (l_k + u_k)/2$ its target center. HAPR computes a composite phase angle for each dimension-pair $j \in \{1, \ldots, B\}$ by aggregating the salted phase shifts of the discrete attributes and the dimension-dependent angular shifts of the numeric attributes. The unified rotation matrix is constructed as:

$$\Phi(\mathbf{t}) = \bigoplus_{j=1}^{B} \mathbf{R}\left(\sum_{m=1}^{M} s_m(j)\psi_{m,j}(t_m^{\text{disc}}) + \sum_{k=1}^{C} \phi_{k,j}^{\text{num}}(t_k^{\text{num}}) \right) \quad . \tag{8}$$

where $\phi_{k,j}^{\text{num}}(a) = \pi \frac{j}{B} \frac{a}{L_k}$ extends to the k-th numeric attribute. Let $\delta_{m,j}^{\text{disc}}$ and $\delta_{k,j}^{\text{num}}$ denote the relative phase differences for the m-th discrete and k-th numeric attributes, respectively. The total relative phase shift between a baked data vector and hybrid query becomes: $\Delta_j = \sum_{m=1}^{M} s_m(j)\delta_{m,j}^{\text{disc}} + \sum_{k=1}^{C} \delta_{k,j}^{\text{num}}$. When all ranges match ($\delta_{k,j}^{\text{num}} = 0$), Δ_j reduces to the discrete sum and Theorem 1 applies as in §4.1; when mismatches, the deterministic shift $\delta_{k,j}^{\text{num}}$ contributes a bounded bias controlled by SPR's in-window bound of §4.2.

Example 5. *Revisit the $d = 8$ space ($B = 4$ pairs) with one exact-match attribute and one numeric attribute over $[0, 100]$. Let the data vector $\mathbf{o}$ have $a^{disc} = 42$ and $a^{num} = 55$, and the query $\mathbf{q}$ satisfy $\langle \mathbf{q}, \mathbf{o} \rangle = 1$. (1) Full match. Query 42 and $[50, 60]$ (center 55): both phase differences vanish, so the rotation is the identity and $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = 1$. (2) Exact-match violation. Query 77 and $[50, 60]$: the numeric center matches, but the discrete offset injects random phases across all 4 pairs, collapsing the score toward zero (Theorem 1). (3) Numeric deviation. Query 42 and $[10, 20]$ (center 15): the discrete attribute aligns, but the numeric value deviates by 40, giving $\Delta_j = \pi \frac{j}{10}$ and $\Delta_4 = 0.4\pi$, attenuating the score to ≈ 0.66 (cf. the identical setting in §4.2). When both mismatch (e.g., 77 and $[10, 20]$), the two perturbations superpose, and the score is no higher than under either mismatch alone.*

4.4 Limitations

We extend the base PART method with advanced mechanisms with APR, SPR, and HAPR. As summarized in Table 3, PART systematically broadens its capability to support complex hybrid filtering conditions (e.g., multi-attribute AND and continuous numeric ranges). This unified geometric framework ensures rigorous multidimensional filtering without sacrificing fundamental search accuracy. Nevertheless, two practical limitations remain for discussion below:

Remark I. PART relies on an offline baking process that embeds attributes into the vector. This inherently dictates that the schema of filtering attributes must be pre-defined; PART cannot support on-the-fly, ad-hoc filtering for arbitrary attributes that were not explicitly baked in advance. In scenarios with highly dynamic attribute schemas, PART is best utilized as a primary filter—baking the most critical or frequently queried attributes (e.g., category, tenant ID, or timestamp) to structurally prune the majority of the search space. Furthermore, PART can be seamlessly incorporated into low-overhead index structures, such as quantization, to facilitate FANNS rapid pruning, which we detail in §5.1.

Remark II. PART does not natively support disjunctive predicates such as OR (including IN as a special case). A viable workaround is query-side splitting: by normalizing the condition into Disjunctive Normal Form (DNF), we evaluate each conjunctive subquery and union the deduplicated results. Although this expansion scales poorly for disjunctions, EQ, multi-attribute AND, and numeric ranges already sufficiently cover most real-world FANNS workloads [20, 27, 28, 53], leaving native OR optimization for future work.

Remark III. PART assumes vectors are *normalized* and exhibit *balanced per-subspace variance*. **(1)** If normalization is relaxed, $\Phi(a)$ still preserves matched IPs, while the unnormalized mismatch IP scales by the known factor $\|\mathbf{o}_r\|\|\mathbf{q}_r\|$. We resolve this by ℓ_2-normalizing vectors prior to baking, and reconstructing the true distances from the retained original norms. **(2)** If the balanced variance assumption is relaxed, the term $\text{Var} = \frac{1}{2}\sum_j w_j$ grows when $\sum_j w_j$ exceeds $1/B$, which degrades the Hoeffding bound $\exp\big(-2\epsilon^2/(4\sum_j w_j)\big)$ and weakens the pruning power. PART mitigates this by employing the random matrix $\mathbf{P}$ of RaBitQ, which equalizes the per-pair variance. Both are detailed in §5.1.

Table 3: Predicates handled by PART.

Predicate	Formal Definition	Guarantee
EQ	$a = t$	mismatch IP $O(1/\sqrt{d})$
AND	$a_1 = t_1 \wedge \cdots \wedge a_M = t_M$	mismatch IP $O(1/\sqrt{d})$
Range	$l \le a \le u$, $a \in [0, L]$	IP deviation $O((w/L)^2)$
Hybrid	$a^{\text{disc}}{=}t^{\text{disc}} \wedge l{\le}a^{\text{num}}{\le}u$	per component
OR	DNF over attributes	per branch

5 IMPLEMENTATIONS OF PART

5.1 RaBitQ Integration

To achieve high efficiency in large-scale ANNS, VQ is indispensable for reducing memory footprint and accelerating distance computations. PART seamlessly integrates with the state-of-the-art VQ algorithm, RaBitQ, leveraging their profound architectural compatibility, which is rooted in three key properties:

- Both architectures rely on orthogonal transformations. RaBitQ utilizes a random orthogonal matrix $\mathbf{P}$ to equalize dimensional variances, satisfying the key requirement of PART for balanced pairwise variance across subspaces. This compatibility guarantees that the mismatch IPs are strictly bounded by $O(1/\sqrt{d})$.
- The attribute rotation $\mathbf{\Phi}(a)$ of PART and the projection matrix $\mathbf{P}$ of RaBitQ are mathematically fusible into a unified operator $\mathbf{P}^\top\mathbf{\Phi}(a)$. Instead of materializing rotation matrices for every attribute, PART only stores a single lightweight seed per attribute value. This renders the additional storage negligible compared to the baseline footprint of $\mathbf{P}$ in RaBitQ. During encoding, the fused operator can be rapidly reconstructed on the fly.
- Because PART strictly preserves vector norms, the integrated method retains the inherent capability of RaBitQ to accurately reconstruct L_2 distances directly from estimated IPs.

We next detail the three core steps of this integrated pipeline.

Offline Baked Binary Quantization. The offline index construction of RaBitQ-PART proceeds in three sequential operations: vector preprocessing, fused vector rotation, and binary quantization. **(1)** Given a raw data vector $\mathbf{o}_r$, we first apply ℓ_2-normalization to obtain the unit vector $\mathbf{o} = \mathbf{o}_r/\|\mathbf{o}_r\|_2$. **(2)** For the normalized vector $\mathbf{o}$ and its attribute value a, PART maps a to its rotation matrix $\Phi(a)$ and fuses it with the shared random orthogonal matrix $\mathbf{P} \in \mathbb{R}^{d\times d}$ of RaBitQ into a single operator $\mathbf{P}^\top\Phi(a)$; only the B phase angles of $\Phi(a)$ are stored per attribute value. **(3)** Binary quantization assigns the vector to its nearest codeword in the base codebook $C = \{\pm 1/\sqrt{d}\}^d$. The optimal codeword is obtained by taking the element-wise sign of the projected vector and rescaling by $1/\sqrt{d}$: $\bar{\mathbf{x}} = \frac{1}{\sqrt{d}}\,\text{sign}\big(\mathbf{P}^\top\Phi(a)\mathbf{o}\big) \in \big\{\pm\frac{1}{\sqrt{d}}\big\}^d$, where $\text{sign}(0) \triangleq +1$.

Online Distance Computation. During the online retrieval phase, PART evaluates the similarity of the baked vectors. Because the

inner products of attribute mismatches are bounded by $O(1/\sqrt{d})$ under the rotated representation, RaBitQ-PART evaluates the continuous distance for all candidates without explicit conditional filtering branches. **(1)** Distance Estimation: The original squared Euclidean distance is reconstructed via the standard geometric expansion. Substituting the IP estimate derived from the baked vectors together with the original norms yields: $\|\mathbf{o}_i - \mathbf{q}\|_2^2 \approx \|\mathbf{o}_i\|_2^2 + \|\mathbf{q}\|_2^2 - 2\widehat{\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle}$. Because the estimation error is bounded by $O(1/\sqrt{d})$, the recovered distances preserve the ranking of valid matches within the theoretical error bounds. **(2)** Hardware Implementation: PART inherits the SIMD-accelerated bitwise operations of RaBitQ. The IP estimate is computed as $\widehat{\langle \hat{\mathbf{q}}, \hat{\mathbf{o}}_i \rangle} = \frac{\langle \bar{\mathbf{x}}_i, \mathbf{P}^\top \hat{\mathbf{q}} \rangle}{c_i}$, where c_i is the normalization factor of the binary code (as in RaBitQ) and $\bar{\mathbf{x}}_i$ is the binary code of $\mathbf{o}_i$. This branch-free bitwise execution unifies attribute filtering and distance approximation into a single scan, improving throughput.

5.2 Index Integration

Because PART maps discrete attribute logic into continuous vector geometry, it operates as a transparent coordinate transformation. This allows PART to seamlessly integrate with standard ANNS indexes, including both IVF and graph-based indexes. We design two deployment paradigms to support different system requirements: building dedicated indexes on the baked space, and executing predicate-agnostic routing on the raw vector space.

Paradigm I: Indexing on the Baked Space. For workloads dominated by filtered queries, the most straightforward approach is to replace the original data vectors with their PART-baked counterparts $\hat{\mathbf{o}}$ during index construction. Because PART preserves distances for identically attributed vectors while ensuring the IPs of mismatched vectors decay to $O(1/\sqrt{d})$, vectors sharing the same attributes naturally cluster together in the high-dimensional baked space. Standard graph indexes like HNSW and NSG can be constructed directly on this baked space without any modification to their routing logic. Furthermore, for quantization-supported frameworks such as IVF-RaBitQ and SymphonyQG, the system can directly ingest the RaBitQ-PART binary codes. During retrieval, the index leverages the hardware-friendly bitwise estimator to execute joint distance evaluation and attribute filtering in a single branch-free scan, aggressively accelerating search throughput.

Paradigm II: Indexing on the Raw Space. To support mixed workloads of both pure ANNS and FANNS, PART supports execution over graph or IVF constructed purely on the raw vectors. When a filtered query is issued, the graph traversal (or cluster probing) relies on the original vector neighborhood structure to navigate toward the target region. During the node evaluation phase, the system intercepts the standard distance computation and injects the RaBitQ-PART two-stage estimator: **(1)** Fast Bitwise Pruning: Using the pre-computed binary codes, PART evaluates the baked IP on the fly. Attribute-mismatched candidates yield IPs of $O(1/\sqrt{d})$ and are pruned from the queue, while valid matches retain accurate estimated distances. **(2)** Post-Retrieval Reranking: The top-k candidates surviving the bitwise filter undergo a lightweight reranking step using the raw vectors and exact attributes. This step resolves the soft attenuation of range predicates and calibrates the final ranking. The primary advantage of this raw-space integration is its zero-duplication memory efficiency. A single raw graph or IVF index seamlessly serves both standard similarity search and complex filtered queries, with PART acting purely as a high-speed, hardware-accelerated evaluator during the candidate assessment phase.

Table 4: Vector dataset statistics.

Vector Dataset	Dimension	Data Size	Query Size	Type
SIFT	128	1,000,000	10,000	Image
MS MARCO	1024	1,000,000	1,000	Text
OpenAI	1536	999,000	1,000	Text
ArXiv	4096	1,000,000	10,000	Text

Table 5: Attribute workload statistics.

Workload	Predicate	Pass Rate	Type
A1_C2	$a_1 = t_1$	50%	Categorical
A1_C10	$a_1 = t_1$	10%	Categorical
A2_C10	$(a_1 = t_1) \wedge (a_2 = t_2)$	1%	Categorical
A3_C10	$\bigwedge_{j=1}^{3}(a_j = t_j)$	0.1%	Categorical
A4_C10	$\bigwedge_{j=1}^{4}(a_j = t_j)$	0.01%	Categorical
ArXiv_Attr	$\bigwedge_{j=1}^{11}(a_j = t_j)$	$\approx$ 30%	Categorical
A1_W50	$\ell_1 \le r_1 \le u_1$	50%	Numeric
A1_W10	$\ell_1 \le r_1 \le u_1$	10%	Numeric
A2_W10	$\bigwedge_{j=1}^{2}(\ell_j \le r_j \le u_j)$	1%	Numeric
A3_W10	$\bigwedge_{j=1}^{3}(\ell_j \le r_j \le u_j)$	0.1%	Numeric
A2_H10	$(a = t) \wedge (\ell \le r \le u)$	1%	Hybrid

6 EXPERIMENTAL EVALUATION

We evaluate the following aspects of PART: (1) search performance of equality, continuous numeric range, and hybrid filters (**§6.2**), (2) index cost (**§6.3**), (3) ablation study (**§6.5**), (4) parameter sensitivity (**§6.6**), and (5) empirical validation of theoretical bounds (**§6.7**). Kindly refer to Appendix for additional evaluations [32].

6.1 Experimental Settings

Datasets. We evaluate PART on four public ANNS datasets: SIFT[2], MS MARCO[3], OpenAI[4], and ArXiv[5]. These datasets span a wide range of dimensionality from 128 to 4096. Table 4 summarizes their dimensionality, database size, query set size, and data modalities.

For every vector dataset, we construct equivalent attribute workloads including equality, continuous numeric ranges, and mixed filters, as summarized in Table 5. The equality workload assigns each vector M categorical attributes, each sampled independently and uniformly from a domain of cardinality C, so that a random query matches a random vector with probability C^{-M}; the expected pass rate over queries is therefore C^{-M}. We instantiate five equality workloads, specifically A1_C2, A1_C10, A2_C10, A3_C10, and A4_C10, where the name Am_Cx denotes an m-way attribute conjunction over domains of cardinality x, covering pass rates from 50% down to 0.01%. These attribute streams are generated independently of the vector datasets. This setting prevents any method from exploiting alignments between similarity neighborhoods and

[2] http://corpus-texmex.irisa.fr/
[3] https://huggingface.co/datasets/CohereLabs/msmarco-v2.1-embed-english-v3
[4] https://huggingface.co/datasets/Qdrant/dbpedia-entities-openai3-text-embedding-3-large-1536-1M
[5] https://hf.co/datasets/SPCL/arxiv-for-fanns-large

attribute values. We similarly construct continuous numeric range workloads over $[0, L]$ with $L = 100$, where Wx denotes a window width of $x\%$ of L (i.e., $w = x$). Under uniformly distributed numeric values, such a window covers exactly $x\%$ of the domain, so its pass rate is $x\%$; we therefore choose the widths to match the pass rates of the equality workloads (50%, 10%, 1%, 0.1%).

Finally, ArXiv [20] serves as the dataset shipping with a real attribute, where each object carries a ArXiv sub category label. We extract this categorical attribute set and apply it to the remaining three vector datasets as an additional workload, yielding an average pass rate of approximately 30%. Unlike synthetic generation, the pass rate is governed by a *highly skewed* category distribution.

Algorithms. We compare PART against prominent IVF-based and graph-based FANNS methods as follows: **(1) IVF-RaBitQ PostFilter** and **(2) IVF-RaBitQ InFilter** serve as baselines derived from IVF-RaBitQ [12]. The former evaluates constraints after vector similarity search, whereas the latter evaluates them on the fly for each candidate during the search process. **(3) FAISS-IVFPQ** [23] executes predicate agnostic IVF scans using an external ID selector, relying on PQ for compressed distance evaluation. **(4) RangePQ** [57] augments a shared PQ index with attribute ordered trees, specializing exclusively in dynamic continuous numeric ranges. **(5) iRangeGraph** [54] specializes in numeric ranges by building a graph hierarchy over segment-tree partitions and dynamically composing query-specific subgraphs on the fly. **(6) SymphonyQG InFilter** serves as a controlled baseline derived from SymphonyQG [15], evaluating constraints on the fly for each candidate during its quantization aware graph traversal. **(7) ACORN** [41] is a predicate agnostic HNSW variant that dynamically executes predicate subgraph traversals to support equality, range, and Boolean filters. **(8) NHQ** [49] builds a composite proximity graph fusing vector and attribute distances, specializing strictly in conjunctive equality over fixed categorical attributes. **(9) FDiskANN** [13] incorporates discrete labels into Vamana construction for label aware pruning, natively supporting only single categorical equality. Our **(10) IVF-RaBitQ PART** and **(11) SymphonyQG PART** natively bake attribute constraints into the quantization space. By integrating the phase rotation framework into both IVFs and SymphonyQG [15], these proposed implementations support equality, multi-attribute `AND`, ranges, and hybrid filters.

Crucially, IVF-RaBitQ PostFilter, IVF-RaBitQ InFilter and SymphonyQG InFilter serve as our apples-to-apples baselines to isolate the performance gain of the PART rotation from the underlying bitwise acceleration.

Metrics and Experimental Setup. We report Recall@100 (Equation 1), computed against the ground-truth top-100 among predicate-satisfying vectors $\mathcal{G}_\phi$, and throughput in QPS. Index size and build time are reported separately in §6.3. All methods are implemented in C++ for both index construction and search, and evaluated on a server with dual AMD EPYC 9534 processors. Index construction uses 48 threads for all methods, while search runs single-threaded per query. We report the average of three independent trials.

Parameter Setting. We detail the parameter configurations below. **(1)** For IVF-based methods, both the RaBitQ variant and the FAISS-IVFPQ baseline share a coarse quantizer with `nlist` = 4096. All RaBitQ variants encode each vector with a 1 bit code and a query refinement $B_{\text{query}} = 4$. In contrast, FAISS-IVFPQ uses $M = D/4$ for all datasets, allocating 8 bits per subspace. To measure the performance tradeoff between recall and QPS, we sweep the search budget by varying `nprobe` from 16 to 1500. **(2)** For graph based methods, the RNG-based methods, including FDiskANN, iRangeGraph, and SymphonyQG, all maintain a maximum degree $R = 32$. The NHQ baseline is built on KGraph with a maximum degree $R = 100$. To establish the same performance tradeoff, we sweep the search budget by scaling the search list L_{search} up to 1000.

6.2 Overall Performance Evaluation

We present an evaluation comparing the search performance of our proposed PART against state-of-the-art baselines, evaluated through the trade-off between QPS and Recall@100. In the presented trade-off grids, curves positioned closer to the top-right corner indicate superior performance. Figure 3 illustrates the experimental results across four datasets: SIFT, MS MARCO, OpenAI, and ArXiv. For each dataset, we evaluate five distinct categorical workloads with varying pass rates ranging from 50% to 0.1%. Notably, the ArXiv-Attr features a highly skewed attribute distribution.

Superiority of PART across Baselines. Across all tested datasets and index architectures, the methods equipped with our PART consistently lie on or near the upper-right frontier of the trade-off grids. For instance, on the OpenAI dataset, SymphonyQG PART achieves 2×–5× higher QPS than graph-based baselines such as NHQ and FDiskANN at a 90% Recall@100 target. Graph-based methods generally outperform IVF-based ones. Attribute-agnostic graph methods such as ACORN are the weakest, since each hop must traverse many neighbors to maintain navigability. For attribute-dedicated methods, NHQ performs well at low pass rates because its composite distance fuses attribute one-hot differences with vector similarity, strengthening links between same-attribute neighbors; however, as the pass rate drops, same-attribute nodes scatter across the graph, so each node can hardly find same-attribute neighbors and connectivity degrades. FDiskANN behaves oppositely: building per-attribute subgraphs lets it quickly locate a same-attribute entry point and subgraph at low pass rates, but at high pass rates, strict subgraph isolation severs global proximity edges between nodes, hurting search performance. Among IVF-based methods, the Post-filtering baselines FAISS-IVFPQ and IVF-RaBitQ PostFilter are consistently the weakest, since they must retrieve a large candidate set before filtering, incurring substantial redundant computation. IVF-RaBitQ InFilter, a Joint-filtering method, consistently outperforms Post-filtering. By feeding attribute-baked vectors directly into IVF-RaBitQ, IVF-RaBitQ PART substantially improves over both its InFilter and PostFilter counterparts.

Performance under Different Pass Rate. At less restrictive pass rates (50% and 10%), standard filtering approaches can usually locate enough valid candidates, so the performance of FANNS algorithms is dominated by the nearest-neighbor navigation capability of the underlying index; most baselines quickly reach over 90% recall, and SymphonyQG InFilter is the best among the baselines, achieving search performance close to SymphonyQG PART. As the pass rate decreases, however, graph indexes designed for vector similarity search—such as SymphonyQG InFilter—degrade rapidly; at a 0.1% pass rate, only FDiskANN retains usable search performance.

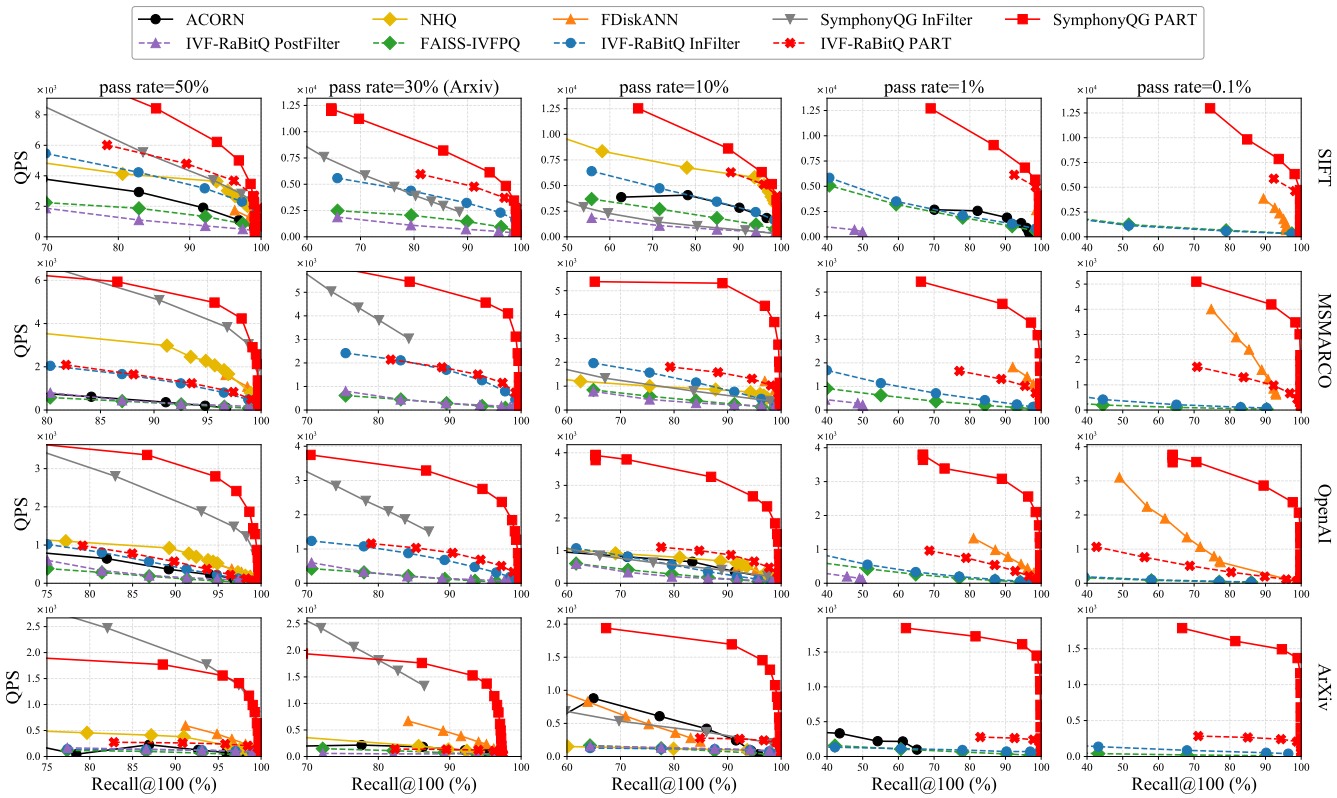


Figure 3: Recall-QPS trade-offs on four vector datasets (rows), five equality workloads with different pass rates (columns). Each point is a search-effort configuration. A curve absent from a panel has no measured point within the displayed recall range.

In contrast, SymphonyQG PART and IVF-RaBitQ PART maintain robust and superior search efficiency across all pass rates.

Robustness under Skewed Attribute Distributions. The second column of Figure 3 compares PART with the baselines under a skewed attribute distribution. Although the pass rate is around 30%, most baselines degrade markedly. We attribute this to the long-tailed nature of the ArXiv attribute: in our 1M sample, values 1 and 2 of `number_of_sub_categories` account for roughly 85% of the base vectors, while values 8–13 contain only 46 vectors in total, and the query labels cover only 1–7, so no query ever targets values 8–13. Consequently, the ground-truth set is itself incomplete: some true neighbors of a query lie in the full ArXiv corpus but are absent from the 1M sample, so no method can recall them. This caps the achievable Recall@100 at approximately 97%, which is a sampling limit, not a method's failure. SymphonyQG PART approaches this ceiling quickly and effectively, whereas other methods suffer a sharp drop in search efficiency. Attribute-dedicated indexes such as FDiskANN cannot navigate into the tiny (or empty) subgraphs of long-tail values; attribute-agnostic indexes such as ACORN waste distance computations on sparse, scattered valid candidates; and post-filtering baselines must retrieve a huge candidate set to accumulate enough long-tail matches. This shows that, by representing attributes as composite phase angles with theoretical guarantees and fusing them directly into the original vectors, PART is largely insensitive to the attribute distribution, whereas existing heuristic methods struggle to guarantee stable performance under skew.

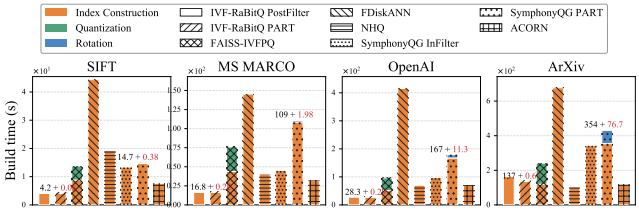


Figure 4: Construction time.

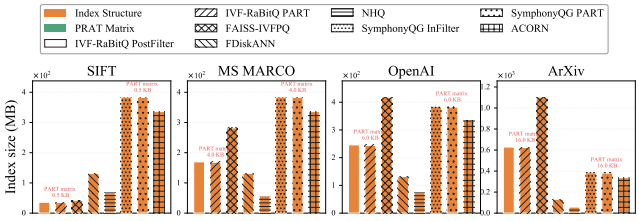


Figure 5: Index size.

6.3 Index Cost

We analyze PART's index overhead in terms of construction time and index size. For index size, generic methods (e.g., ACORN) report their exact size, whereas for attribute-dedicated methods (e.g., NHQ) the size is averaged over different attribute workloads. Since

Table 6: QPS at 90% Recall@100 for range filtered queries. "—" indicates that the method does not reach the target recall.

Pass Rate	Method	QPS at Recall@100 = 90%		
		SIFT	MS MARCO	OpenAI
0.1%	PART	**848**	**152**	**61**
	FAISS	469	35	20
1%	PART	**2195**	**595**	**219**
	FAISS	1153	308	54
10%	PART	**2839**	**1214**	**587**
	FAISS	1494	528	186

IVF-RaBitQ PostFilter and IVF-RaBitQ InFilter share the same underlying index and differ only in when filtering is applied, we report the index cost once, under IVF-RaBitQ PostFilter.

Construction Time. Figure 4 reports the offline construction overhead of PART. As the build-time breakdown shows, the PART-specific rotation accounts for only a marginal fraction of total index construction time. On SIFT and MS MARCO, the rotation overhead of IVF-RaBitQ PART is negligible (0.08s and 0.22s); even on the 4096-dimensional, skewed ArXiv dataset, the rotation in SymphonyQG PART takes 76.7s, against 354s for base graph construction. This low overhead follows from PART's design: block-diagonal orthogonal matrices are reconstructed on the fly from globally shared salts. By unifying attributes and vectors at the representation level, PART keeps both throughput and construction cost low.

Index Size. Figure 5 reports the offline index storage overhead of PART across four datasets. Both the inverted-index-based IVF-RaBitQ PART and the graph-based SymphonyQG PART remain nearly identical in size to their non-attribute-fused baselines, IVF-RaBitQ PostFilter and SymphonyQG InFilter. This is because PART stores only the phase information needed for the orthogonal transformation: the core PART matrix adds only 0.5 KB, 4.0 KB, 6.0 KB, and 16.0 KB on SIFT, MS MARCO, OpenAI, and ArXiv, respectively. Compared to base indexes consuming hundreds of MB or even a gigabyte—such as SymphonyQG on ArXiv and the storage-heavy ACORN and FAISS-IVFPQ—this kilobyte-level overhead is negligible. PART thus integrates into mainstream index architectures with near-zero storage redundancy, fusing attributes and vectors without compromising scalability.

6.4 Search Performance for Complex Predicates

Table 6 reports QPS at 90% Recall@100 for range-filtered queries. We compare against FAISS (IVF + PQ with range filtering) because the two range-specialized baselines, iRangeGraph and RangePQ, do not natively support *multi-attribute range AND*. Specifically, iRangeGraph builds one binary-search-tree-partitioned subgraph per range attribute; when C range attributes are conjoined, it would need to materialize the cross-product of C partitionings, neither of which its index supports. RangePQ augments a single PQ index with one attribute-ordered tree, and likewise handles only a single range attribute. We therefore use FAISS as the representative single-range baseline and report results for one, two, and three conjoined range attributes (A1_W10, A2_W10, A3_W10), leaving a native multi-range extension of iRangeGraph/RangePQ to future work.

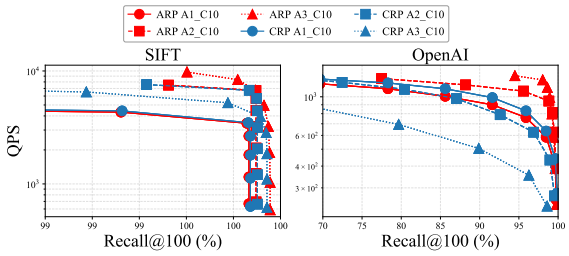


Figure 6: APR vs. CPR on multi-attribute AND workloads.

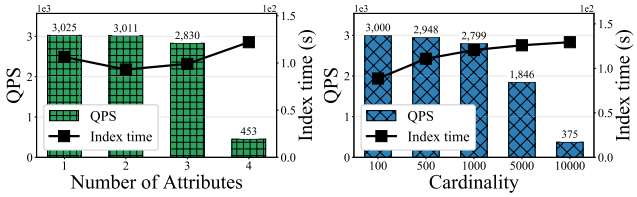


Figure 7: Parameter sensitivity of PART.

6.5 Ablation Study

This section compares two phase-rotation strategies, APR and CPR, on multi-attribute AND workloads (A1_C10, A2_C10, A3_C10) using the SIFT and OpenAI datasets, in terms of Recall@100 and QPS. On the lower-dimensional SIFT dataset, both APR and CPR (red solid vs. blue dashed lines in Figure 6) lie in the upper-right region, reaching 99%–100% recall at 10^3–10^4 QPS, with little difference between them. On the higher-dimensional OpenAI dataset, however, the two strategies diverge: APR remains in the high-recall region (Recall@100 near 100%) while sustaining high throughput, whereas CPR degrades markedly. This shows that, under high-dimensional vectors and complex multi-attribute filters, APR offers substantially more stable and efficient retrieval than CPR.

6.6 Parameter Sensitivity

To evaluate the robustness of PART under different workload conditions, we conduct a parameter sensitivity analysis focusing on two key factors: the number of attributes and the attribute cardinality. All evaluations in this section are conducted on the OpenAI dataset using the SymphonyQG PART. The results, detailing both QPS (measured at a target Recall@100 of 90%) and index time, are presented in Figure 7. In our experiments, we isolate the effect of each parameter by fixing the other: when evaluating the number of queried attributes, the attribute cardinality is fixed at 10; conversely, when evaluating cardinality, the number of attributes is fixed at 1.

Impact of Number of Attributes. Our method satisfies the 90% recall requirement across all feasible multi-attribute conditions supported by the dataset. As the number of attributes increases from 1 to 3, the QPS remains highly stable, with only a marginal degradation from 3025 to 2830. This confirms that our mechanism efficiently handles multi-attribute AND predicate constraints. When the number of attributes reaches 4, the QPS drops to 453. This is not a limitation of our method; instead, it reflects the extreme boundary of the filtering space. Given that the OpenAI dataset contains $N = 999{,}000$ base vectors and the attribute cardinality is fixed at 10, the combined pass rate for four attributes decreases to 10^{-4}. This reduces the expected number of valid candidates across the entire dataset to approximately 100 ($999{,}000 \times 10^{-4} = 99.9$),

which corresponds to the strictest valid query condition for a top-100 search. Notably, the index construction time remains stable under these settings, indicating that encoding additional attributes does not severely impact the offline construction.

Impact of Attribute Cardinality. The right panel shows the effect of attribute cardinality (100 to 10,000). QPS stays high (3,000–2,799) for cardinalities 100–1,000, then drops to 1,846 and 375 at 5,000 and 10,000. This is *not* a storage effect: PART computes phases on the fly from shared salts, keeping its footprint at the KB level regardless of domain size (§6.3). It is a *pass-rate effect*—a single equality attribute over domain C leaves only N/C eligible candidates (≈ 200 and ≈ 100 at $C = 5{,}000$ and 10,000). The latter hits the top-k boundary of §2.2, the same effect behind the four-way AND case. The mild index-time increase comes from encoding more distinct attribute values during ingestion, an $O(1)$-per-vector cost, not from the rotation itself. Overall, PART adapts well to varying cardinalities, maintaining 90% Recall@100 even at 10,000.

6.7 Validation of Theoretical Bounds

To bridge the gap between mathematical limits and real-world execution, we design a series of empirical micro-benchmarks to validate our theorems over actual data distributions. Specifically, we select three datasets with progressively increasing dimensionalities from 1024 to 4096. All vectors are normalized. For each dataset, we inject uniformly distributed categorical attributes. This methodology allows us to trace the empirical distortion between the original IP $\langle \mathbf{q}, \mathbf{o} \rangle$ and the baked IP $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle$ under varying dimensionalities.

Validation of Attribute Matching Soundness. To validate the $1/d$ convergence mechanism of Theorem 1, we track the baked IPs for mismatched vectors across the three datasets. As depicted in Figure 10, the black dashed lines represent the theoretical 3σ bounds ($y = \pm 3/\sqrt{d}$). The vertical spread of the mismatched vectors is overwhelmingly confined within these theoretical limits, achieving containment rates of 99.9%, 99.6%, and 99.9% on MS MARCO, OpenAI, and ArXiv, respectively. Moreover, this mismatch error bound noticeably tightens as the dimensionality d increases from 1024 to 4096. This alignment between the empirical spread and the theoretical limits demonstrates that PART successfully suppresses the baked IPs of attribute-mismatched vectors to near zero. It also confirms that the variance shrinkage is governed by d and inherently agnostic to the underlying data distribution. Given that vectors with near-zero IPs are virtually impossible to emerge as true top-k nearest neighbors in high-dimensional spaces, PART prunes mismatched candidates at the distance-evaluation level.

Validation of Vector Similarity Completeness. We analyze the baked IPs for matched vector pairs across the same datasets. As shown by the blue points in Figure 10, the baked IPs of identically attributed vectors align flawlessly along the $y = x$ diagonal. This zero-distortion phenomenon confirms that $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$ holds universally for all valid matches, regardless of the underlying vector dimensionality or data distribution. From a system perspective, this exact similarity preservation guarantees that the relative distance ranking among the filtered candidates remains entirely unperturbed. Consequently, PART supports NNS without introducing any false negatives or ranking degradation for true matches.

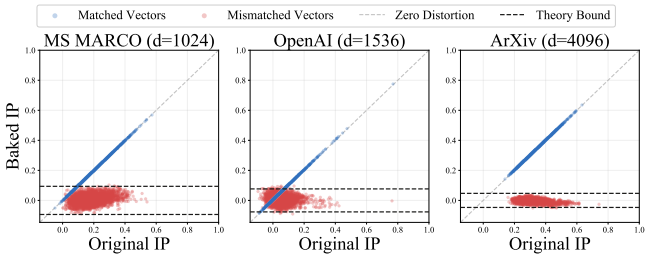


Figure 8: Verification of soundness and completeness.

7 RELATED WORK

FANNS Methods. FANNS relies on execution-level strategies—Pre-, Post-, or Joint-filtering (e.g., FDiskANN [13], NHQ [49], ACORN [41], iRangeGraph [54])—which treat the predicate as a discrete test external to distance computation, limiting pruning; no single strategy dominates across pass rates [28]. Pass-rate-adaptive workarounds, CAPS [17], UNIFY [31], FAVOR [43], merely rearrange the external predicate. PART instead bakes attributes into the vector space, removing the predicate from the search loop and achieving high throughput with stable recall across all pass rates.

Beyond pass rate, recent work addresses dynamic maintenance and arbitrary filtering. Methods that fuse predicates into the index topology (SeRF [60], iRangeGraph [54]) require costly reconstruction on updates and support a single predicate type; dynamic range indices (DIGRA [21], RangePQ [57], WoW [51]) support incremental updates but rely on auxiliary structures. PART runs on general-purpose indices via RaBitQ-PART, matching attributes and estimating distances with error guarantees; updates require only re-encoding the quantized codes, with no index reconstruction. It supports multi-attribute AND, ranges, and hybrid predicates.

Skewed attribute distributions (e.g., the heavy-tailed cardinalities in ArXiv [20]) further challenge existing methods: heuristic joint-filtering lacks hard guarantees and can suffer recall collapses, while mitigations such as graph re-balancing (TFANNS [35]) and auto-tuning (PGTuner [8]) are distribution-specific. PART embeds constraints into vector coordinates via angular rotations and, backed by theoretical guarantees, keeps accuracy and matching stable under skew without distribution-specific structures or tuning.

Rotation-Based Encoding. Encoding a discrete signal as a phase rotation evaluated via a plain IP—the primitive underlying PART—is well established in adjacent fields: RoPE [44] represents token positions as complex-plane rotations, and RotatE [46] models knowledge-graph relations as rotations. Rotations are also prevalent in VQ: OPQ aligns vector spaces with quantization subspaces [37], and RaBitQ [12] applies a random projection before one-bit sign quantization to achieve a distribution-independent $O(1/\sqrt{d})$ error bound. PART builds on RaBitQ by inserting an attribute-driven rotation *between* its preconditioning and sign quantization, repurposing the orthogonal transform for double duty: decorrelating coordinates for quantization and embedding the filtering semantics.

8 CONCLUSION

In this paper, we proposed PART, a method that resolves the low pass rate dilemma in FANNS. By baking discrete attribute constraints into high-dimensional vectors via orthogonal phase rotations, PART translates predicate matching into vector distance computation with theoretical error guarantees. This design can be naturally extended to support complex query conditions, including multi-attribute AND semantics, numerical range filtering, and hybrid predicates. Furthermore, it serves as a drop-in frontend that can be readily integrated with mainstream ANNS architectures such as VQ, IVF, and proximity graphs. Extensive evaluations across diverse workloads, including heavily skewed real-world datasets, demonstrate that PART achieves state-of-the-art query throughput across the entire pass rates spectrum while maintaining recall targets.

REFERENCES

[1] Anas Ait Aomar, Karima Echihabi, Marco Arnaboldi, Ioannis Alagiannis, Damien Hilloulin, and Manal Cherkaoui. 2025. RWalks: Random Walks as Attribute Diffusers for Filtered Vector Search. *Proc. ACM Manag. Data* 3, 3, Article 212 (June 2025), 26 pages. https://doi.org/10.1145/3725349

[2] Martin Aumüller, Erik Bernhardsson, and Alexander Faithfull. 2020. ANN-Benchmarks: A benchmarking tool for approximate nearest neighbor algorithms. *Inf. Syst.* 87, C (Jan. 2020), 13. https://doi.org/10.1016/j.is.2019.02.006

[3] Artem Babenko and Victor Lempitsky. 2012. The inverted multi-index. In *2012 IEEE Conference on Computer Vision and Pattern Recognition*. 3069–3076. https://doi.org/10.1109/CVPR.2012.6248038

[4] Helena Caminal, Yannis Chronis, Yannis Papakonstantinou, Fatma Özcan, and Anastasia Ailamaki. 2025. Filtered Vector Search: State-of-the-art and Research Challenges. *Proceedings of the VLDB Endowment* 18, 12 (2025), 5488–5492. https://doi.org/10.14778/3750601.3750700

[5] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. 2021. SPANN: Highly-efficient Billion-scale Approximate Nearest Neighborhood Search. In *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (Eds.), Vol. 34. Curran Associates, Inc., 5199–5212. https://proceedings.neurips.cc/paper_files/paper/2021/file/299dc35e747eb77177d9cea10a802da2-Paper.pdf

[6] Paul Covington, Jay Adams, and Emre Sargin. 2016. Deep Neural Networks for YouTube Recommendations. In *Proceedings of the 10th ACM Conference on Recommender Systems* (Boston, Massachusetts, USA) *(RecSys '16)*. Association for Computing Machinery, New York, NY, USA, 191–198. https://doi.org/10.1145/2959100.2959190

[7] Sanjoy Dasgupta and Yoav Freund. 2008. Random projection trees and low dimensional manifolds. In *Proceedings of the Fortieth Annual ACM Symposium on Theory of Computing* (Victoria, British Columbia, Canada) *(STOC '08)*. Association for Computing Machinery, New York, NY, USA, 537–546. https://doi.org/10.1145/1374376.1374452

[8] Hao Duan, Yitong Song, Bin Yao, and Anqi Liang. 2025. PGTuner: An Efficient Framework for Automatic and Transferable Configuration Tuning of Proximity Graphs. *Proc. ACM Manag. Data* 3, 4, Article 261 (Sept. 2025), 27 pages. https://doi.org/10.1145/3749179

[9] Casper Benjamin Freksen. 2021. An introduction to johnson-lindenstrauss transforms. *arXiv preprint arXiv:2103.00564* (2021).

[10] Cong Fu, Changxu Wang, and Deng Cai. 2022. High Dimensional Similarity Search With Satellite System Graph: Efficiency, Scalability, and Unindexed Query Compatibility. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 8 (2022), 4139–4150. https://doi.org/10.1109/TPAMI.2021.3067706

[11] Cong Fu, Chao Xiang, Changxu Wang, and Deng Cai. 2019. Fast approximate nearest neighbor search with the navigating spreading-out graph. *Proc. VLDB Endow.* 12, 5 (Jan. 2019), 461–474. https://doi.org/10.14778/3303753.3303754

[12] Jianyang Gao and Cheng Long. 2024. RaBitQ: Quantizing High-Dimensional Vectors with a Theoretical Error Bound for Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 2, 3, Article 167 (May 2024), 27 pages. https://doi.org/10.1145/3654970

[13] Siddharth Gollapudi, Neel Karia, Varun Krishnaswamy, et al. 2024. Filtered-DiskANN: Graph Algorithms for Approximate Nearest Neighbor Search with Filters. In *Proceedings of the ACM Web Conference (WWW)*. https://doi.org/10.1145/3543507.3583552

[14] Long Gong, Huayi Wang, Mitsunori Ogihara, and Jun Xu. 2020. iDEC: indexable distance estimating codes for approximate nearest neighbor search. *Proc. VLDB Endow.* 13, 9 (May 2020), 1483–1497. https://doi.org/10.14778/3397230.3397243

[15] Yutong Gou, Jianyang Gao, Yuexuan Xu, and Cheng Long. 2025. Symphonyqg: Towards symphonious integration of quantization and graph for approximate nearest neighbor search. *Proceedings of the ACM on Management of Data* 3, 1 (2025), 1–26.

[16] Mihajlo Grbovic and Haibin Cheng. 2018. Real-time Personalization using Embeddings for Search Ranking at Airbnb. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (London, United Kingdom) *(KDD '18)*. Association for Computing Machinery, New York, NY, USA, 311–320. https://doi.org/10.1145/3219819.3219885

[17] Gaurav Gupta, Jonah Yi, Benjamin Coleman, Chen Luo, Vihan Lakshman, and Anshumali Shrivastava. 2023. Caps: A practical partition index for filtered similarity search. *arXiv preprint arXiv:2308.15014* (2023).

[18] Jui-Ting Huang, Ashish Sharma, Shuying Sun, Li Xia, David Zhang, Philip Pronin, Janani Padmanabhan, Giuseppe Ottaviano, and Linjun Yang. 2020. Embedding-based Retrieval in Facebook Search. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (Virtual Event, CA, USA) *(KDD '20)*. Association for Computing Machinery, New York, NY, USA, 2553–2561. https://doi.org/10.1145/3394486.3403305

[19] Qiang Huang, Jianlin Feng, Yikai Zhang, Qiong Fang, and Wilfred Ng. 2015. Query-aware locality-sensitive hashing for approximate nearest neighbor search. *Proc. VLDB Endow.* 9, 1 (Sept. 2015), 1–12. https://doi.org/10.14778/2850469.2850470

[20] Patrick Iff, Paul Brügger, Marcin Chrapek, David Kochergin, Maciej Besta, and Torsten Hoefler. 2026. Benchmarking Filtered Approximate Nearest Neighbor Search Algorithms on Transformer-based Embedding Vectors. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Melbourne, VIC, Australia) *(SIGIR '26)*. Association for Computing Machinery, New York, NY, USA, 610–622. https://doi.org/10.1145/3805712.3809731

[21] Mengxu Jiang, Zhi Yang, Fangyuan Zhang, Guanhao Hou, Jieming Shi, Wenchao Zhou, Feifei Li, and Sibo Wang. 2025. DIGRA: A Dynamic Graph Indexing for Approximate Nearest Neighbor Search with Range Filter. *Proc. ACM Manag. Data* 3, 3, Article 148 (June 2025), 26 pages. https://doi.org/10.1145/3725399

[22] Yicheng Jin, Yongji Wu, Wenjun Hu, Bruce Maggs, Jun Yang, Xiao Zhang, and Danyang Zhuo. 2026. Curator: Efficient Vector Search with Low-Selectivity Filters. *Proc. ACM Manag. Data* 4, 1, Article 21 (April 2026), 27 pages. https://doi.org/10.1145/3786635

[23] Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2019. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data* 7, 3, 535–547. https://doi.org/10.1109/TBD.2019.2917033

[24] Herve Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product Quantization for Nearest Neighbor Search. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 33, 1 (2011), 117–128. https://doi.org/10.1109/TPAMI.2010.57

[25] Jinhyuk Lee, Feiyang Chen, Sahil Dua, Daniel Cer, Madhuri Shanbhogue, Iftekhar Naim, Gustavo Hernández Ábrego, Zhe Li, Kaifeng Chen, Henrique Schechter Vera, et al. 2025. Gemini embedding: Generalizable embeddings from gemini. *arXiv preprint arXiv:2503.07891* (2025).

[26] Patrick Lewis, Ethan Pérez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[27] Mocheng Li, Baotong Lu, James Cheng, and Chenhao Ma. 2026. EMA: Approximate Nearest Neighbor Search with General Attribute Filtering and Dynamic Updates. *arXiv preprint arXiv:2606.00734* (2026).

[28] Mocheng Li, Xiao Yan, Baotong Lu, Yue Zhang, James Cheng, and Chenhao Ma. 2025. Attribute Filtering in Approximate Nearest Neighbor Search: An In-depth Experimental Study. *Proc. ACM Manag. Data* 3, 6, Article 298 (Dec. 2025), 26 pages. https://doi.org/10.1145/3769763

[29] Mingjie Li, Ying Zhang, Yifang Sun, Wei Wang, Ivor W. Tsang, and Xuemin Lin. 2020. I/O Efficient Approximate Nearest Neighbour Search based on Learned Functions. In *2020 IEEE 36th International Conference on Data Engineering (ICDE)*. 289–300. https://doi.org/10.1109/ICDE48307.2020.00032

[30] Zhaoheng Li, Silu Huang, Wei Ding, Yongjoo Park, and Jianjun Chen. 2025. SIEVE: Effective Filtered Vector Search with Collection of Indexes. *Proc. VLDB Endow.* 18, 11 (July 2025), 4723–4736. https://doi.org/10.14778/3749646.3749725

[31] Anqi Liang, Pengcheng Zhang, Bin Yao, Zhongpu Chen, Yitong Song, and Guangxu Cheng. 2024. UNIFY: Unified Index for Range Filtered Approximate Nearest Neighbors Search. *Proc. VLDB Endow.* 18, 4 (Dec. 2024), 1118–1130. https://doi.org/10.14778/3717755.3717770

[32] Lsyhprum. 2025. *PART: Technical Report*. Technical Report. GitHub. https://github.com/Lsyhprum/PART/blob/main/technical_report.pdf Contains detailed proofs, derivations, and supplementary experimental results.

[33] Duo Lu, Helena Caminal, Manos Chatzakis, Yannis Papakonstantinou, Yannis Chronis, Vaibhav Jain, and Fatma Özcan. 2026. An In-Depth Study of Filter-Agnostic Vector Search on a PostgreSQL Database System: [Experiments & Analysis]. *Proc. ACM Manag. Data* 4, 3, Article 134 (May 2026), 26 pages. https://doi.org/10.1145/3802011

[34] Kejing Lu, Hongya Wang, Wei Wang, and Mineichi Kudo. 2020. VHP: approximate nearest neighbor search via virtual hypersphere partitioning. *Proc. VLDB Endow.* 13, 9 (May 2020), 1443–1455. https://doi.org/10.14778/3397230.3397240

[35] Jiarui Luo, Miao Qiao, Chaoji Zuo, and Dong Deng. 2025. Tag-Filtered Approximate Nearest Neighbor Search. In *2025 IEEE 41st International Conference on Data Engineering (ICDE)*. 3642–3654. https://doi.org/10.1109/ICDE65448.2025.00272

[36] Yu A. Malkov and D. A. Yashunin. 2020. Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 4 (2020), 824–836. https://doi.org/10.1109/TPAMI.2018.2889473

[37] Yusuke Matsui, Yusuke Uchida, Hervé Jégou, and Shin'ichi Satoh. 2018. A survey of product quantization. *ITE Transactions on Media Technology and Applications* 6, 1 (2018), 2–10.

[38] Marius Muja and David G. Lowe. 2014. Scalable Nearest Neighbor Algorithms for High Dimensional Data. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 36, 11 (2014), 2227–2240. https://doi.org/10.1109/TPAMI.2014.2321376

[39] James Jie Pan, Jianguo Wang, and Guoliang Li. 2024. Survey of vector database management systems. *The VLDB Journal* 33, 5 (July 2024), 1591–1615. https://doi.org/10.1007/s00778-024-00864-x

[40] Omiros Papaspiliopoulos. 2020. High-dimensional probability: An introduction with applications in data science.

[41] Liana Patel, Peter Kraft, Carlos Guestrin, and Matei Zaharia. 2024. ACORN: Performant and Predicate-Agnostic Search Over Vector Embeddings and Structured Data. *Proceedings of the ACM on Management of Data (PACMMOD)* 2, 3. https://doi.org/10.1145/3654923

[42] Gaurav Sehgal and Semih Salihoglu. 2025. NaviX: A Native Vector Index Design for Graph DBMSs With Robust Predicate-Agnostic Search Performance. *Proceedings of the VLDB Endowment* 18, 11. https://doi.org/10.14778/3749646.3749704

[43] Junjie Song, Yu Liu, Guoyu Hu, Zhongle Xie, Ming Yang, Beng Chin Ooi, and Ke Zhou. 2026. FAVOR: Efficient Filter-Agnostic Vector ANNS Based on Selectivity-Aware Exclusion Distances. *Proceedings of the ACM on Management of Data* 4, 3 (2026), 1–25. https://doi.org/10.1145/3802060

[44] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing* 568 (2024), 127063. https://doi.org/10.1016/j.neucom.2023.127063

[45] Suhas Jayaram Subramanya, Devvrit, Rohan Kadekodi, Ravishankar Krishnaswamy, and Harsha Vardhan Simhadri. 2019. Randomized Algorithms for Fast Nearest Neighbor Search. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[46] Zhiqing Sun, Zhi-Hong Deng, Jian-Yun Nie, and Jian Tang. 2019. Rotate: Knowledge graph embedding by relational rotation in complex space. *arXiv preprint arXiv:1902.10197* (2019).

[47] Jianguo Wang, Xiaomeng Yi, Rentong Guo, Haijin Jin, Peng Xu, Shengjun Li, Xueqiang Wang, et al. 2021. Milvus: A Purpose-Built Vector Data Management System. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*. https://doi.org/10.1145/3448016.3457550

[48] Jianguo Wang, Xiaomeng Yi, Rentong Guo, Hai Jin, Peng Xu, Shengjun Li, Xiangyu Wang, Xiangzhou Guo, Chengming Li, Xiaohai Xu, Kun Yu, Yuxing Yuan, Yinghao Zou, Jiquan Long, Yudong Cai, Zhenxiang Li, Zhifeng Zhang, Yihua Mo, Jun Gu, Ruiyi Jiang, Yi Wei, and Charles Xie. 2021. Milvus: A Purpose-Built Vector Data Management System. In *Proceedings of the 2021 International Conference on Management of Data* (Virtual Event, China) *(SIGMOD '21)*. Association for Computing Machinery, New York, NY, USA, 2614–2627. https://doi.org/10.1145/3448016.3457550

[49] Mengzhao Wang, Lingwei Lv, Xiaoliang Xu, Yuxiang Wang, Qiang Yue, and Jiongkang Ni. 2023. An efficient and robust framework for approximate nearest neighbor search with attribute constraint. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 692, 14 pages.

[50] Mengzhao Wang, Xiaoliang Xu, Qiang Yue, and Yuxiang Wang. 2021. A comprehensive survey and experimental comparison of graph-based approximate nearest neighbor search. *Proc. VLDB Endow.* 14, 11 (July 2021), 1964–1978. https://doi.org/10.14778/3476249.3476255

[51] Ziqi Wang, Jingzhe Zhang, and Wei Hu. 2025. WoW: A Window-to-Window Incremental Index for Range-Filtering Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 3, 6, Article 378 (Dec. 2025), 27 pages. https://doi.org/10.1145/3769843

[52] Chuangxian Wei, Bin Wu, Sheng Wang, Renjie Lou, Chaoqun Zhan, Feifei Li, and Yuanzhe Cai. [n.d.]. AnalyticDB-V: A Hybrid Analytical Engine Towards Query Fusion for Structured and Unstructured Data. *Proc. VLDB Endow.* 13, 12 ([n. d.]), 3152–3165.

[53] Qianqian Xiong and Mengxuan Zhang. 2026. Query-aware Routing for Filtered Approximate Nearest Neighbors Search. *arXiv preprint arXiv:2606.19898* (2026).

[54] Yuexuan Xu, Jianyang Gao, Yutong Gou, Cheng Long, and Christian S. Jensen. 2024. iRangeGraph: Improvising Range-dedicated Graphs for Range-filtering Nearest Neighbor Search. *Proceedings of the ACM on Management of Data (PACMMOD)* 2, 6. https://doi.org/10.1145/3698814

[55] Qiang Yue, Mengzhao Wang, Xiaoliang Xu, Cheng Long, Yuxiang Wang, and Jiahui Wang. 2026. Efficient and Robust Out-Of-Distribution Vector Similarity Search with Cross-Distribution Monotonic Graph. *Proc. ACM Manag. Data* 4, 1, Article 29 (April 2026), 29 pages. https://doi.org/10.1145/3786643

[56] Fangyuan Zhang, Mengxu Jiang, Guanhao Hou, Jieming Shi, Hua Fan, Wenchao Zhou, Feifei Li, and Sibo Wang. 2025. Efficient Dynamic Indexing for Range Filtered Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 3, 3, Article 152 (June 2025), 26 pages. https://doi.org/10.1145/3725401

[57] Fangyuan Zhang, Mengxu Jiang, Guanhao Hou, Jieming Shi, Hua Fan, Wenchao Zhou, Feifei Li, and Sibo Wang. 2025. Efficient Dynamic Indexing for Range Filtered Approximate Nearest Neighbor Search. *Proceedings of the ACM on Management of Data* 3, 3 (2025), 1–26. https://doi.org/10.1145/3725401

[58] Shuai Zhang, Lina Yao, Aixin Sun, and Yi Tay. 2019. Deep Learning Based Recommender System: A Survey and New Perspectives. *ACM Comput. Surv.* 52, 1, Article 5 (Feb. 2019), 38 pages. https://doi.org/10.1145/3285029

[59] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, et al. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176* (2025).

[60] Chaoji Zuo, Miao Qiao, Wenchao Zhou, Feifei Li, and Dong Deng. 2024. SeRF: Segment Graph for Range-Filtering Approximate Nearest Neighbor Search. *Proceedings of the ACM on Management of Data (PACMMOD)* 2, 1. https://doi.org/10.1145/3639324

A APPENDIX

A.1 Theoretical Analysis of CPR

Theorem 3 (CPR Completeness and Vulnerability). *Assume a multi-attribute query targeting* $\mathbf{t} = (t_1, \ldots, t_M)$*. Under Composite Phase Rotation (CPR) with M uniform blocks, if a normalized data vector* $\mathbf{o}$ *matches all attributes (*$\mathbf{a} = \mathbf{t}$*), the baked IP is preserved:* $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$*. However, if exactly one attribute mismatches (e.g.,* $a_1 \neq t_1$ *and* $a_m = t_m$ *for* $m \geq 2$*), and assuming uniform variance distribution across blocks, the expected baked IP degrades to* $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = \frac{M-1}{M} \langle \mathbf{q}, \mathbf{o} \rangle$*, violating the zero-mean soundness.*

Proof. Let the dimension pairs be partitioned into M disjoint blocks $\{\mathcal{B}_1, \ldots, \mathcal{B}_M\}$. The baked IP decomposes into a sum over these blocks: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \sum_{m=1}^{M} (\Phi_{\mathcal{B}_m}(t_m)\mathbf{q}^{(m)})^\top (\Phi_{\mathcal{B}_m}(a_m)\mathbf{o}^{(m)})$. When $\mathbf{a} = \mathbf{t}$, for all m, $\Phi_{\mathcal{B}_m}(t_m)^\top \Phi_{\mathcal{B}_m}(a_m) = \mathbf{I}$, giving $\sum_{m=1}^{M} \langle \mathbf{q}^{(m)}, \mathbf{o}^{(m)} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$. When exactly one attribute mismatches (e.g., $a_1 \neq t_1$), the rotation on block $\mathcal{B}_1$ yields a random phase shift, while the rotations on the remaining $M-1$ blocks act as identity matrices. Thus: $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = X_1 + \sum_{m=2}^{M} \langle \mathbf{q}^{(m)}, \mathbf{o}^{(m)} \rangle$, where X_1 is the baked IP of the mismatched block. Since $\mathbb{E}[X_1] = 0$ (following Theorem 1 over $\mathcal{B}_1$), the expected total IP evaluates strictly to the sum of the matched blocks. Under uniform variance assumption, $\sum_{m=2}^{M} \langle \mathbf{q}^{(m)}, \mathbf{o}^{(m)} \rangle = \frac{M-1}{M} \langle \mathbf{q}, \mathbf{o} \rangle$. The false-positive score is bounded only by the reduced dimensionality d/M, degrading the mismatch penalty. □

A.2 Theoretical Analysis of APR

Theorem 4 (APR Completeness and Soundness). *Assume a multi-attribute query targeting* $\mathbf{t} = (t_1, \ldots, t_M)$*. Under Additive Phase Rotation (APR), if a normalized data vector* $\mathbf{o}$ *matches all attributes (*$\mathbf{a} = \mathbf{t}$*), the baked IP is strictly preserved:* $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$*. If any attribute mismatches (*$\mathbf{a} \neq \mathbf{t}$*), the baked IP guarantees zero mean,* $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$*, and restores the full-dimensional variance bound* $\mathrm{Var}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 1/d$*, effectively suppressing false positives.*

Proof. Let $\Delta_j = \sum_{m=1}^{M} s_m(j)\big(\psi_{m,j}(a_m) - \psi_{m,j}(t_m)\big)$ be the total relative phase shift on the j-th dimension pair. When $\mathbf{a} = \mathbf{t}$, $\psi_{m,j}(a_m) - \psi_{m,j}(t_m) = 0$ for all m and j. Thus, $\Delta_j = 0$ across all $B = d/2$ pairs. The overall relative rotation $\Phi(\mathbf{t})^\top \Phi(\mathbf{a}) = \mathbf{I}$, strictly preserving $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \langle \mathbf{q}, \mathbf{o} \rangle$. When $\mathbf{a} \neq \mathbf{t}$, suppose there exists at least one mismatched attribute m' (i.e., $a_{m'} \neq t_{m'}$). The total phase shift on pair j is $\Delta_j = s_{m'}(j)\delta_{m',j} + \sum_{m \neq m'} s_m(j)\delta_{m,j}$. Because the hash yields a random phase difference $\delta_{m',j}$, and the salt $s_{m'}(j)$ is drawn independently for every pair j, the resulting shift Δ_j behaves as an independent random variable uniformly distributed in $[0, 2\pi)$ across *all* $d/2$ dimension pairs. Consequently, the superposition scrambles the entire vector space globally. The scenario mathematically reduces to the single-attribute mismatch case defined in Theorem 1, yielding $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$ and $\mathrm{Var} = 1/d$. □

A.3 Theoretical Analysis of SPR

Theorem 5 (SPR Bounded Preservation and Attenuation). *Assume a continuous numeric attribute* $a \in [0, L]$ *and a range query* $[l, u]$ *with width* $w = u - l$ *and center* $t = (l + u)/2$*. Under Scaled Phase Rotation (SPR), assuming matched query and data vectors: **(1)** In-Window Preservation: If* $a \in [l, u]$*, the baked IP deviation is strictly bounded by* $O((w/L)^2)$*. By partitioning the query into m equal-width sub-queries, this error bound compresses to* $O((w/(mL))^2)$*, guaranteeing exact IP convergence. **(2)** Out-of-Window Attenuation: If* $a \notin [l, u]$*, the phase shifts grow linearly with the distance* $|a - t|$ *and the dimension index ratio* j/B*, predictably attenuating the baked IP for out-of-window candidates.*

Proof. The relative phase shift on the j-th dimension pair is $\Delta_j = \phi_j(a) - \phi_j(t) = \pi \frac{j}{B} \frac{a-t}{L}$. Assuming uniform variance distribution across dimension pairs, the expected baked IP is mathematically equivalent to $\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle = \frac{1}{B} \sum_{j=1}^{B} \langle \mathbf{q}^{(j)}, \mathbf{o}^{(j)} \rangle \cos \Delta_j$.

For **(1)**: When $a \in [l, u]$, the distance to the center is bounded by $|a - t| \leq w/2$. Thus, the maximum phase shift on any pair is $|\Delta_j| \leq \pi \frac{j}{B} \frac{w}{2L} \leq \frac{\pi w}{2L}$. For small Δ_j, the Maclaurin series expansion gives $\cos \Delta_j \approx 1 - \Delta_j^2/2$. The total inner product deviation is therefore proportional to $\frac{1}{B} \sum_{j=1}^{B} \frac{\Delta_j^2}{2} \leq \frac{1}{2}\left(\frac{\pi w}{2L}\right)^2 = O((w/L)^2)$. When partitioning the range into m sub-queries, the effective width of each target center reduces to w/m. Substituting this into the bound yields a maximum deviation of $O((w/(mL))^2)$, ensuring that in-window similarities are preserved with arbitrarily high precision as m increases. For **(2)**: When $a \notin [l, u]$, $|a - t| > w/2$. The angular shift Δ_j diverges from 0, growing proportionally with the numeric violation $|a - t|$. Furthermore, the scaling factor j/B acts as a frequency multiplier, ensuring that higher-index dimension pairs experience aggressive rotation. This deterministically shrinks $\cos \Delta_j$, penalizing the final aggregated inner product to exclude out-of-window vectors. □

A.4 Theoretical Analysis of HAPR

Theorem 6 (HAPR Bounded Completeness and Soundness). *Assume a hybrid query targeting M discrete attributes* $\mathbf{t}^{disc}$ *and C numeric range centers* $\mathbf{t}^{num}$*. Under Hybrid Additive Phase Rotation (HAPR): **(1)** Bounded Completeness: If all discrete attributes match and numeric values fall within their respective query windows (width* w_k*), the baked IP deviation is bounded by* $O(\sum_{k=1}^{C} (w_k/L_k)^2)$*. **(2)** Soundness (Discrete Mismatch): If* any *discrete attribute mismatches, the expected baked IP collapses to zero,* $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$*, with variance* $1/d$*, entirely neutralizing false positives regardless of numeric values. **(3)** Deterministic Attenuation (Numeric Mismatch): If discrete attributes match but numeric values deviate out-of-window, the IP reliably attenuates based on the aggregated scaled phase shifts.*

Proof. The total phase shift on the j-th dimension pair is $\Delta_j = \sum_{m=1}^{M} s_m(j)\delta_{m,j}^{\text{disc}} + \sum_{k=1}^{C} \delta_{k,j}^{\text{num}}$. For **(1)** (Match): When $\mathbf{a}^{\text{disc}} = \mathbf{t}^{\text{disc}}$, all discrete shifts vanish ($\delta_{m,j}^{\text{disc}} = 0$). For in-window numeric attributes, the deterministic shift is bounded by $|\delta_{k,j}^{\text{num}}| \leq \pi \frac{w_k}{2L_k}$. Since $\cos(\Delta_j) \approx 1 - \Delta_j^2/2$ for small Δ_j, the total IP deviation is bounded by the squared sum of maximum numeric shifts, yielding $O(\sum_k (w_k/L_k)^2)$. (If $a_k = t_k$ exactly, $\Delta_j = 0$ and IP is strictly preserved). For **(2)** (Discrete Mismatch): Suppose a discrete attribute m' mismatches. The independent salt $s_{m'}(j)$ ensures the term $s_{m'}(j)\delta_{m',j}^{\text{disc}}$ is uniformly distributed in $[0, 2\pi)$. Because the sum of a uniform random variable and any other variables (including the deterministic numeric shifts $\delta_{k,j}^{\text{num}}$) modulo 2π remains strictly uniformly distributed in $[0, 2\pi)$, Δ_j acts as an independent random phase across

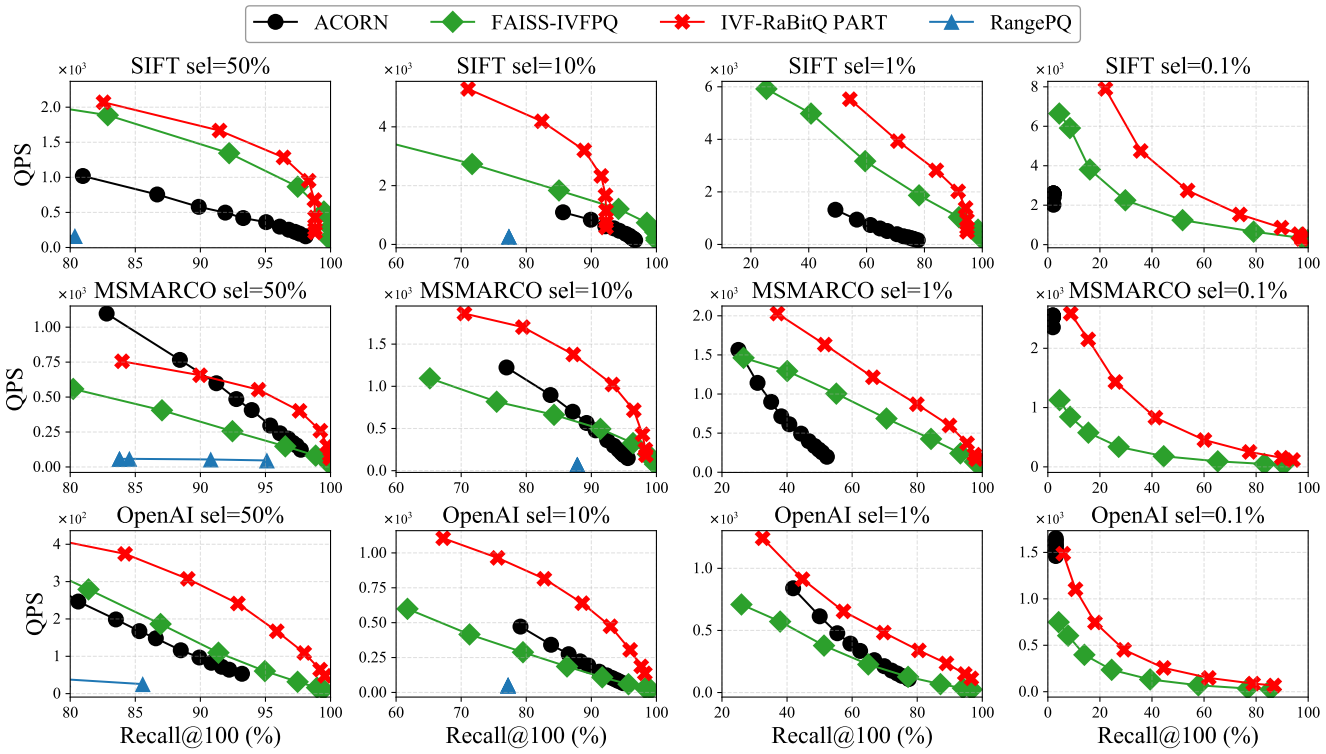


Figure 9: Performance of numeric range filtering.

all $B = d/2$ pairs. The formulation mathematically reduces to Theorem 1, yielding $\mathbb{E}[\langle \hat{\mathbf{q}}, \hat{\mathbf{o}} \rangle] = 0$ and $\text{Var} = 1/d$. For **(3)** (Numeric Mismatch): When discrete attributes match but numeric ones deviate, $\Delta_j = \sum_{k=1}^{C} \pi \frac{j}{B} \frac{a_k - t_k}{L_k}$. The deterministic angular offset grows linearly with the coordinate distance from the range centers, safely suppressing out-of-window candidates as established in SPR. □

A.5 Performance of Numeric Range Filtering

We evaluate the QPS versus Recall@100 trade-off for IVF-RaBitQ PART, RangePQ, ACORN, and FAISS-IVFPQ under continuous numeric range queries across the SIFT, MSMARCO, and OpenAI datasets, with the pass rate scaling down from 50% to an extreme 0.1%. Across all evaluated datasets and pass rate settings, our proposed IVF-RaBitQ PART (marked by red crosses) consistently dominates the upper-right Pareto frontier, demonstrating its ability to sustain superior query throughput at identical recall targets compared to all baselines. As the pass rate approaches 0.1%, the robustness disparity among the underlying architectures becomes severely magnified. Notably, ACORN (black circles) suffers a catastrophic performance collapse under extreme pass rates, with its data points entirely collapsing to the extreme left of the axes, indicating that its graph topology fails to achieve high recall when valid candidates are exceedingly sparse. In contrast, both IVF-RaBitQ PART and the overall suboptimal FAISS-IVFPQ (green diamonds) exhibit high resilience, extending smoothly and robustly toward the 100% recall boundary. Finally, RangePQ (blue triangles) emerges as the weakest baseline; it is barely visible near the bottom of the y-axis at 50% and 10% pass rates and fails to register effectively at stricter pass rates, proving that its filtering mechanism incurs prohibitive overhead and abysmal query efficiency under such workloads.

A.6 Performance on Hybrid Predicates

We evaluate the QPS versus Recall@100 trade-off for IVF-RaBitQ PART, ACORN, and FAISS-IVFPQ under hybrid filtering workloads across the SIFT, MS MARCO, and OpenAI datasets. Across all evaluated datasets, our proposed IVF-RaBitQ PART overwhelmingly dominates the upper-right Pareto frontier, sustaining exceptionally high query throughput while smoothly approaching the 100% recall boundary. In contrast, while the predicate-agnostic graph index ACORN manages to reach high recall regimes, it incurs a massive performance penalty, yielding QPS values up to an order of magnitude lower than IVF-RaBitQ PART. This severe degradation exposes the fundamental topological vulnerability of existing graph routing mechanisms when navigating complex, multi-dimensional hybrid constraints. Finally, FAISS-IVFPQ exhibits the weakest performance; it not only delivers the lowest overall throughput but also suffers a steep, cliff-like collapse between 80% and 90% recall across all three datasets, completely failing to extend into the high-recall spectrum. These results firmly validate the superior efficiency and structural robustness of the PART architecture when jointly evaluating discrete and continuous hybrid predicates.

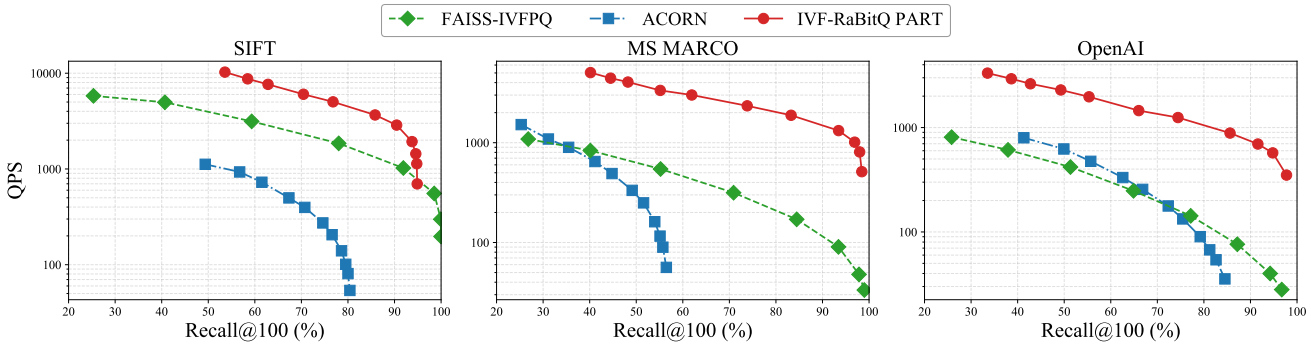


Figure 10: Performance on hybrid predicates.